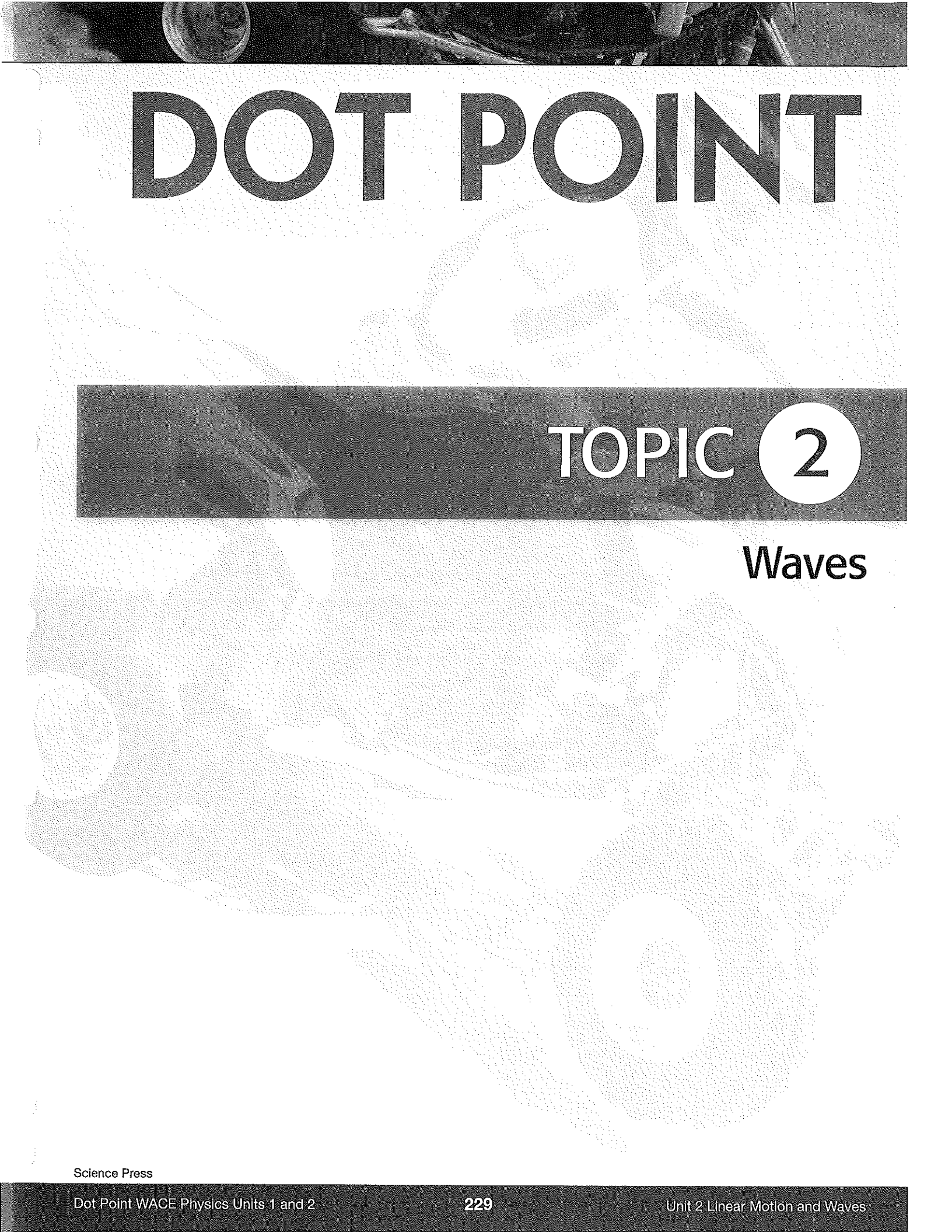




DOT POINT

TOPIC 2

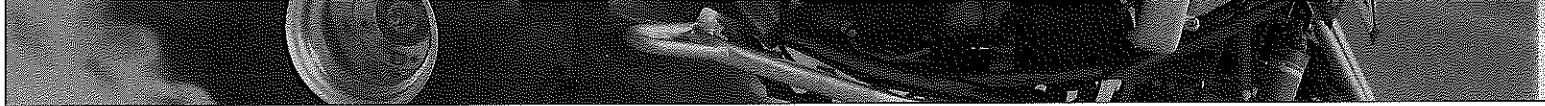
Waves

2.1 Longitudinal and transverse waves.

2.1.1 Properties of waves.

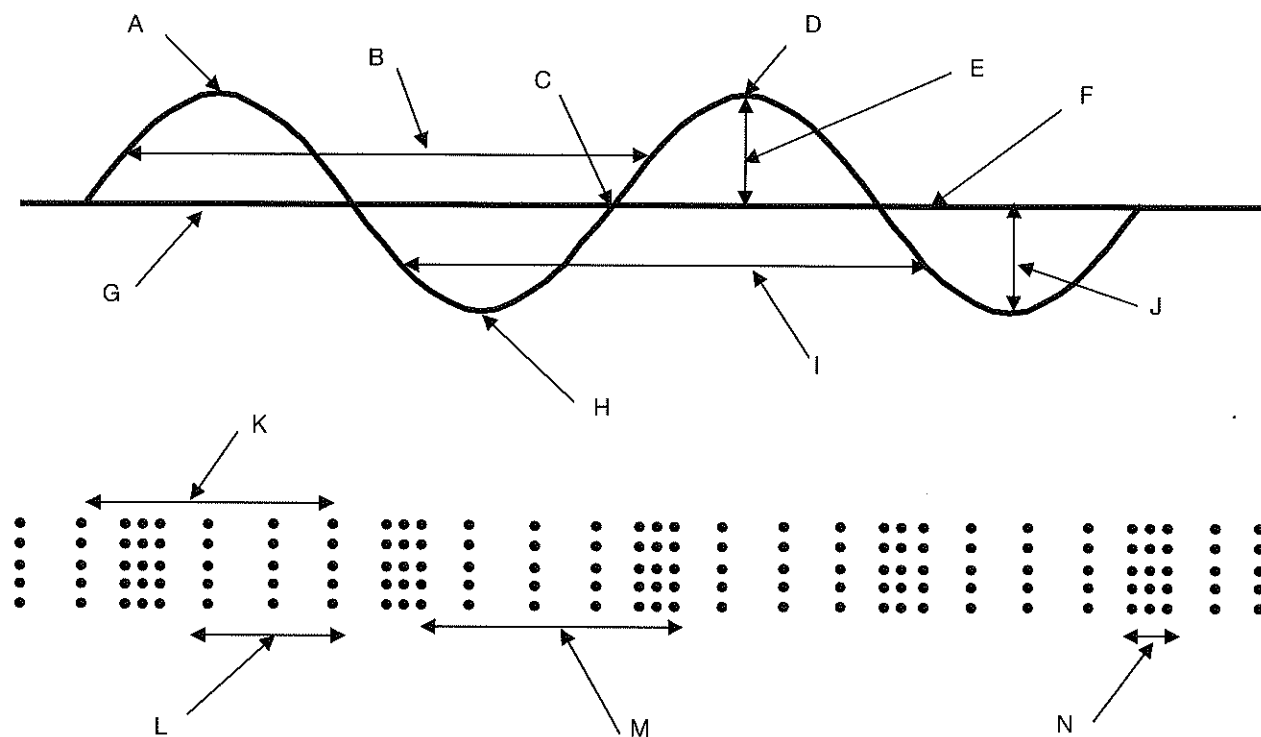
2.1.1.1 All waves have one property in common – they all transfer energy from one place to another. Their other properties are jumbled up in the table. Your task is to unjumble them to find the correct definitions using language syntax or knowledge. Each definition requires a phrase from each column. Write your completed sentences in the answer table provided so you can learn the definitions and properties.

Wave property	Phrase 1	Phrase 2
1. A wave pulse	(a) soundwave determines the pitch (highness)	(A) from its rest or zero displacement position.
2. A wave	(b) is a single energy-carrying disturbance	(B) identical point on the next wave pulse.
3. Crests	(c) time it takes for one complete pulse to	(C) transfers energy from one place to another.
4. Troughs are	(d) of a wave is the distance from a point on a wave pulse to the	(D) this determines the loudness or softness of the sound.
5. The wavelength	(e) lower than normal density	(E) pass a fixed point.
6. The frequency of	(f) the lowest points on transverse waves. They	(F) of waves that pass a fixed position each second.
7. The frequency of a	(g) is the distance from the zero displacement position of the	(G) waves. They are positions of maximum particle displacement.
8. The period of a wave is the	(h) are the highest points on transverse	(H) of the medium carrying a longitudinal wave.
9. Wave velocity is a	(i) of higher than normal	(I) or lowness) of the sound produced.
10. The amplitude of a wave	(j) carrying a wave is a measure of how far a particle is	(J) particles to the top of a crest or the bottom of a trough.
11. The amplitude of a wave determines	(k) is a series of pulses which	(K) density of the medium carrying a longitudinal wave.
12. A compression is a region	(l) how much energy the wave is carrying. For a soundwave	(L) its position when the medium is not transferring energy.
13. A rarefaction is a region of	(m) position of a particle in a medium is	(M) also represent positions of maximum particle displacement.
14. The displacement of a particle in a medium	(n) measure of how fast the wave transfers energy	(N) in a medium or cycle of an electromagnetic wave.
15. The zero displacement	(o) a wave motion is equal to the number	(O) from one place to another.



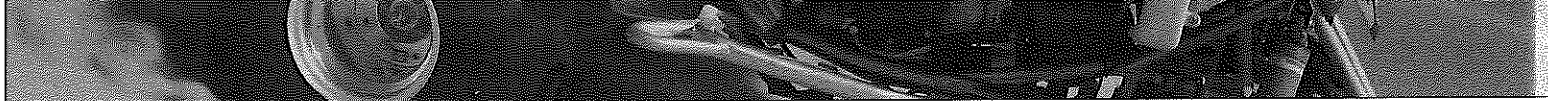
Wave property	Completed sentences
1. A wave pulse	
2. A wave	
3. Crests	
4. Troughs are	
5. The wavelength	
6. The frequency of	
7. The frequency of a	
8. The period of a wave is the	
9. Wave velocity is a	
10. The amplitude of a wave	
11. The amplitude of a wave determines	
12. A compression is a region	
13. A rarefaction is a region of	
14. The displacement of a particle in a medium	
15. The zero displacement	

2.1.1.2 Use your definitions from Question 2.1.1.1 to identify each label on the following diagrams of waves. Write your answers in the table provided.



Position	Property
A	
B	
C	
D	
E	
F	
G	

Position	Property
H	
I	
J	
K	
L	
M	
N	



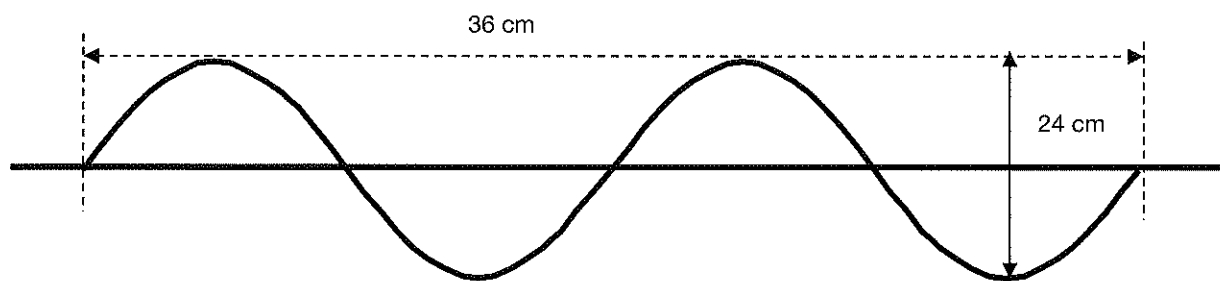
- 2.1.1.3** What is the amplitude of a wave?
(A) The maximum displacement of a particle in the medium from its rest position.
(B) The distance the wave moves in one time period of the wave.
(C) The distance the wave moves in one second.
(D) The distance equal to one wavelength.
- 2.1.1.4** What is the term for the distance from crest of one wave pulse to the crest of the next wave pulse?
(A) Period. (B) Frequency. (C) Amplitude. (D) Wavelength.
- 2.1.1.5** After a wave has passed through a medium, how does the position of the particles of the medium compare to their original positions before the wave enters the medium?
(A) The particles' positions are about the same as their original position.
(B) The particles' positions depend on the type of wave that has passed through the medium.
(C) The particles' positions move in the direction of the wave.
(D) The particles' positions move perpendicular to their original position.
- 2.1.1.6** As a mechanical wave travels through a medium there is a net transfer of what?
(A) Energy only. (B) Mass only.
(C) Both mass and energy. (D) Neither mass nor energy.
- 2.1.1.7** What do mechanical waves transfer?
(A) Energy. (B) Liquid.
(C) Matter. (D) Particles of the medium.
- 2.1.1.8** Hertz is a measure of:
(A) Amplitude. (B) Frequency. (C) Period. (D) Wavelength.
- 2.1.1.9** Which of the following applies to the frequency of a wave?
(A) s (B) s^{-1} (C) m (D) m^{-1}
- 2.1.1.10** A wave that has a relatively high frequency will also have a relatively:
(A) Large amplitude. (B) Low amplitude. (C) Short wavelength. (D) Long period.
- 2.1.1.11** Which of the following affects the speed of the wave most?
(A) Its wavelength. (B) Its frequency.
(C) Its amplitude. (D) The medium it travels in.
- 2.1.1.12** What term describes the number of waves that occur in 1 s?
(A) Amplitude. (B) Frequency. (C) Period. (D) Wavelength.
- 2.1.1.13** What term do we give to the number of cycles of a periodic wave occurring per unit time?
(A) Wavelength. (B) Frequency. (C) Amplitude. (D) Period.
- 2.1.1.14** Which property of a mechanical wave measures the amount of particle vibration?
(A) Amplitude. (B) Period. (C) Frequency. (D) Wavelength.
- 2.1.1.15** Many wave properties are dependent upon other wave properties. Yet, one wave property is independent of all other wave properties. Which one of the following properties of a wave is independent of all the others?
(A) Amplitude. (B) Frequency. (C) Period. (D) Wavelength.
- 2.1.1.16** What happens to the frequency of a wave in a constant medium if its amplitude increases?
(A) It becomes faster. (B) It increases.
(C) It decreases. (D) It remains constant.

2.1.2 Transverse matter waves.

2.1.2.1

- (a) What is the distinguishing characteristic of a transverse matter wave?
- (b) In the space provided, draw a labelled diagram to illustrate your answer to (a).

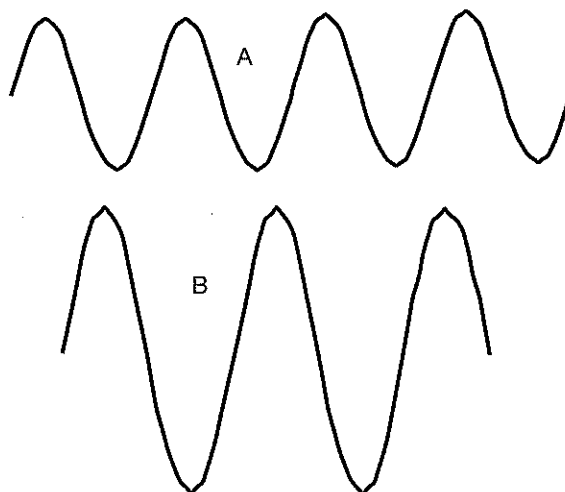
2.1.2.2 Consider this wave. The wave represents a time period of 0.25 s.

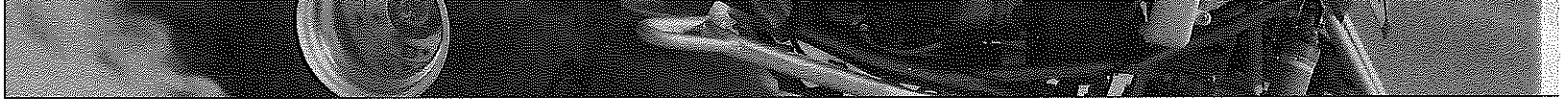


- (a) What is the wavelength of the wave?
- (b) What is its amplitude?
- (c) How many wave pulses are shown in the diagram?
- (d) How many wavelengths are shown in the diagram?
- (e) What is the period of the wave?
- (f) What is its frequency?

2.1.2.3 Each of the two waves shown here represents a time period of 0.5 s. For each, assuming they are drawn life size, determine the wavelength, frequency, period and amplitude.

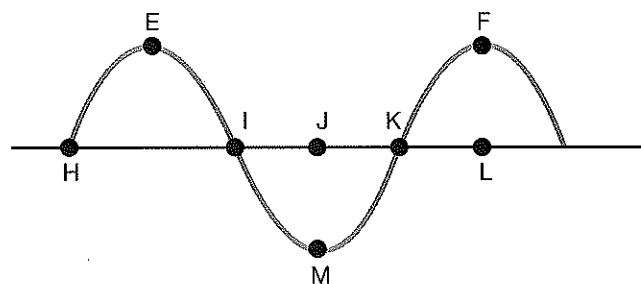
	Wave A	Wave B
Wavelength		
Frequency		
Period		
Amplitude		





2.1.2.4 Consider the transverse wave shown.

- (a) Identify a wavelength.
- (b) Identify an amplitude.
- (c) Identify a crest.
- (d) Identify a trough.
- (e) Identify a particle which has zero displacement on this wave.



2.1.2.5 When asked to define the wavelength of a transverse matter wave, two students gave the following answers.

Student X: The wavelength is the distance from the crest of a pulse on the wave to the crest on the next pulse in the wave.

Student Y: The wavelength is the distance from any point on a pulse in the wave to the identical point on the next pulse in the wave.

Which student's definition is better? Justify your answer.

2.1.2.6 If two transverse matter waves have the same frequency, what also must they have that is the same?

- (A) Amplitude.
- (B) Energy.
- (C) Period.
- (D) Wavelength.

2.1.2.7 The distance between two successive troughs can be labelled as:

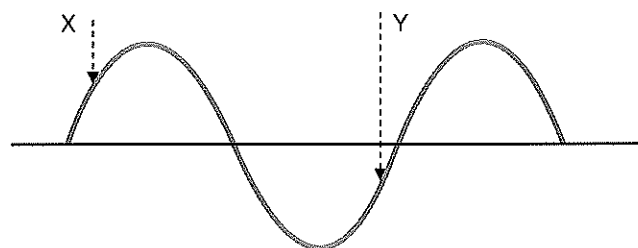
- (A) Amplitude.
- (B) Period.
- (C) Velocity.
- (D) Wavelength.

2.1.2.8 The amount of energy carried by a transverse matter wave is dependent on its:

- (A) Amplitude.
- (B) Period.
- (C) Velocity.
- (D) Wavelength.

2.1.2.9 On the wave shown, which correctly labels the distance between X and Y?

- (A) $\frac{\lambda}{2}$
- (B) $\frac{2\lambda}{3}$
- (C) $\frac{3\lambda}{4}$
- (D) λ

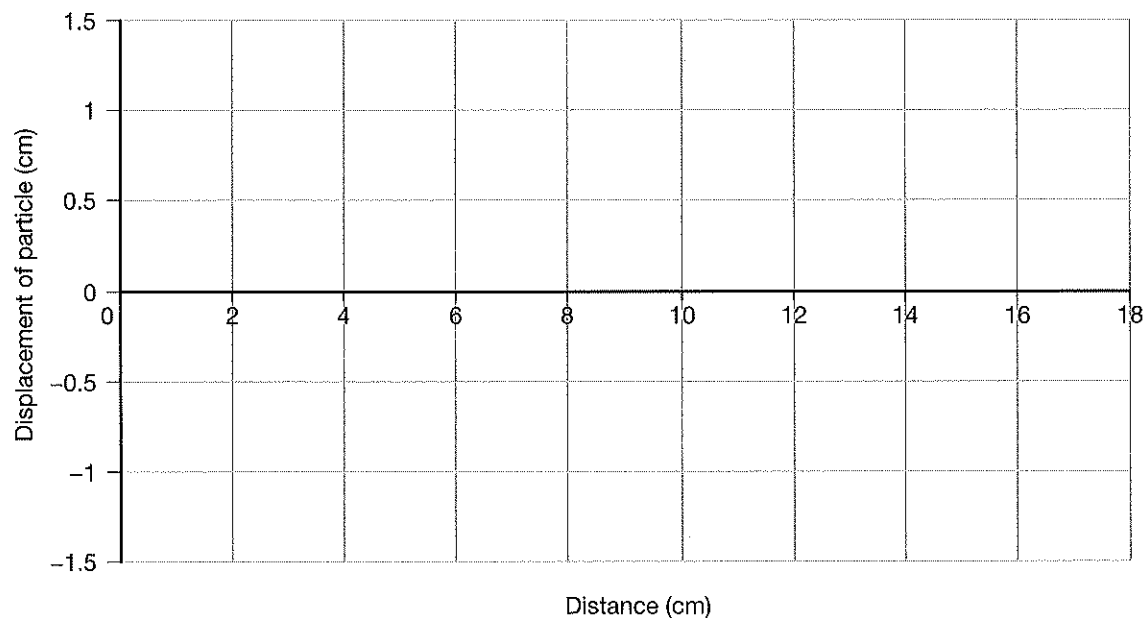


2.1.2.10 Wave X has wavelength λ and frequency f . Wave Y, in the same medium, has wavelength 2λ . What will be its frequency?

- (A) $\frac{f}{2}$
- (B) f
- (C) $2f$
- (D) Unable to be determined.

2.1.2.11

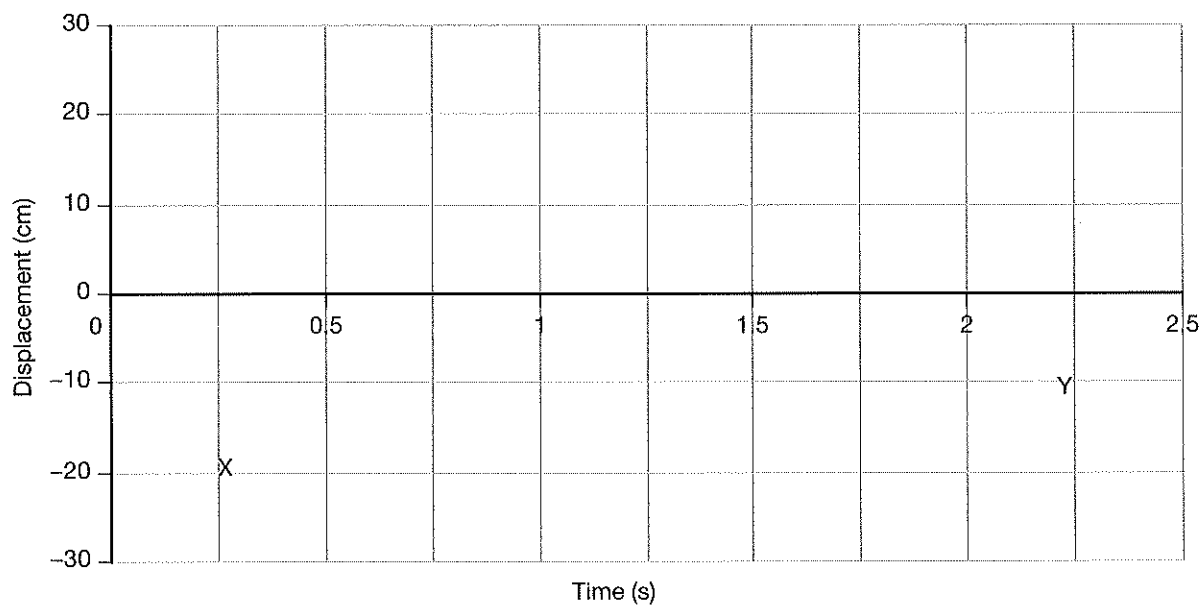
- (a) On the grid, draw three wavelengths of a wave which has an amplitude of 15 mm and a wavelength of 6 cm. Label this graph X.



- (b) On the same axes draw two wavelengths of another graph which has amplitude of 10 mm and a wavelength of 8 cm. Label this graph Y.

2.1.2.12

- (a) On the grid, draw four wavelengths of a wave which has an amplitude of 25 mm and a frequency of 2.0 Hz. Label this graph X.



- (b) On the same axes draw three wavelengths of another graph which has amplitude of 20 mm and a period of 0.75 s. Label this graph Y.



2.1.3 Longitudinal matter waves.

2.1.3.1 The following statements all refer to longitudinal matter waves. Classify each statement as either true (T) or false (F).

☐	(a) Longitudinal waves are mechanical waves.
☐	(b) They pass readily through a vacuum.
☐	(c) Longitudinal waves are produced when there is a source of vibration in a medium such as the vibration of a drum skin, the reed in a wind instrument, a piano string, a human larynx, all of which set the air near them vibrating.
☐	(d) In a longitudinal wave, particles oscillate back and forth at right angles to the direction of energy transfer.
☐	(e) A compression is a region where particles of the medium are closer together than when they are at their rest positions.
☐	(f) A rarefaction is a region where the particles of the medium are further apart than when in their rest positions.
☐	(g) The rest positions of particles are their positions when no energy is being transferred through the medium.
☐	(h) The amplitude of a soundwave determines the loudness or softness of the sound, also known as its pitch.
☐	(i) The larger the amplitude, the louder the sound.
☐	(j) The frequency of a soundwave determines the highness or lowness of the sound, also known as its intensity.
☐	(k) A high frequency wave produces a high pitch sound.
☐	(l) When sound is reflected (bounced back) from a hard surface, we hear an echo.

2.1.3.2 For each of the statements above you classified as false, rewrite them so that they are true.

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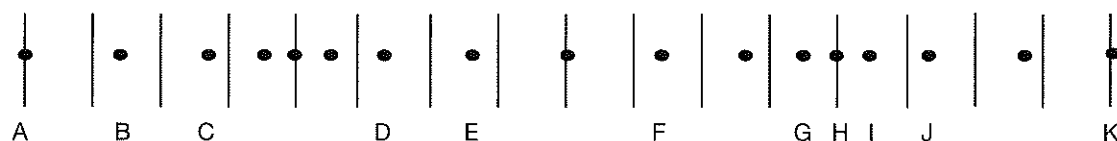
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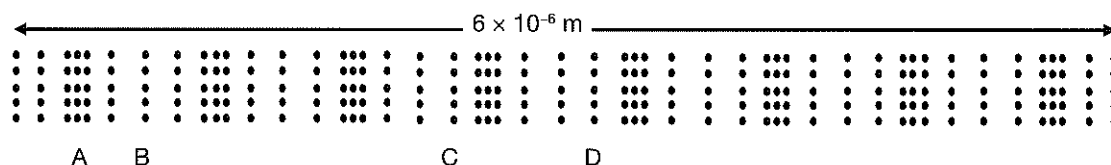
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- 2.1.3.3** Consider the diagram below which represents a longitudinal wave travelling through a medium. Consider the wave to have been drawn life size. The vertical lines represent the rest positions of the matter particles.



- Identify and measure a wavelength on this wave.
- Identify an amplitude on this wave.
- Identify a labelled particle which has zero displacement on this wave.
- Identify a labelled particle which is displaced to the right of its rest position.
- Identify a labelled particle which is displaced to the left of its rest position.
- Identify a zero displacement position which is unoccupied.

- 2.1.3.4** Consider the diagram below which represents a longitudinal wave travelling through a medium.



(a)	How many wave pulses are shown in the diagram?		(f)	What is its frequency?	
(b)	How many wavelengths are shown in the diagram?		(g)	Which way are the particles in the medium for this wave oscillating?	
(c)	What is the distance AB on this diagram?		(h)	If the wave was to carry more energy, what property would be different?	
(d)	What distance is represented by CD?		(i)	If the frequency of this wave was doubled, what would happen to its speed? Explain your answer.	
(e)	If the wave takes 0.6 s to travel from position B to position D, what is the period of the wave?		(j)	Why is it very difficult to determine the amplitude of the wave from a diagram like this?	

2.1.4 Transverse electromagnetic waves.

2.1.4.1 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise the kinetic theory of matter.

Completed sentences	Second halves (not in order)
All electromagnetic waves can	in hertz (Hz).
Electromagnetic waves all travel at the speed of light ($3 \times 10^8 \text{ m s}^{-1}$) in a	wave is indicated by the amplitude of the wave.
Electromagnetic waves are self-propagating alternating	the peaks of successive magnetic or electric field pulses.
Because the motion of the changing magnetic and electric fields are at right angles to the direction in which they carry energy, electromagnetic	vacuum – they slow down a little in other media.
Because electromagnetic waves are really hard to draw, we usually draw	its frequency.
The flaw in doing this is that the energy carried by a transverse	on the number of photons in the beam.
In electromagnetic waves, the energy is directly	travel through a vacuum.
The wavelength of an electromagnetic wave is the distance between	proportional to the frequency of the photons which make up the radiation.
We usually refer to the intensity of an	that pass a point each second.
The intensity of an electromagnetic wave depends	electric and magnetic fields.
Each photon will have energy dependent on	wavelength to pass a given point.
The period of an electromagnetic wave is the time for one	waves are also classified as transverse waves.
The frequency of an electromagnetic wave is the number of wavelengths	electromagnetic wave rather than to its amplitude.
The frequency of electromagnetic waves is measured	them as transverse matter waves.

2.1.4.2 Consider this diagram which represents an electromagnetic wave in the air. The scale of the diagram is $1 \text{ cm} = 10^{-11} \text{ m}$.

(a) What is the distance AB on this diagram?

(b) What distance is represented by CD?

(c) What is represented by the distance EF?

(d) What is the wavelength of the wave?

(e) How many wave pulses are shown in the diagram?

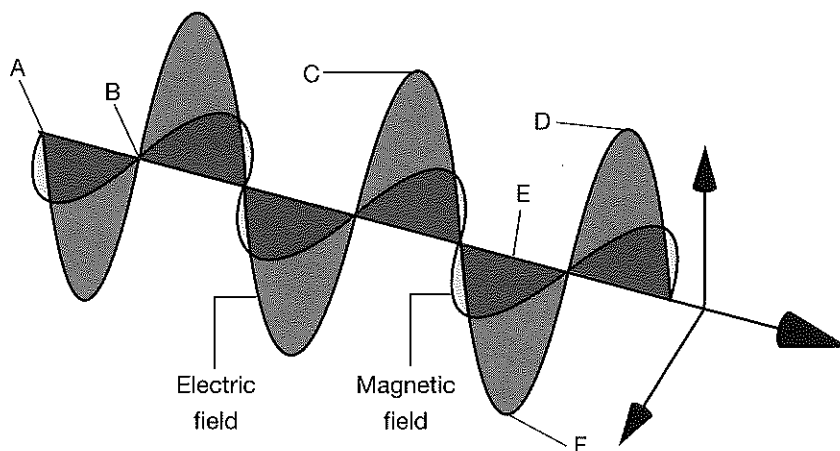
(f) How many wavelengths are shown in the diagram?

(g) What is the speed of this wave in air? Justify your answer.

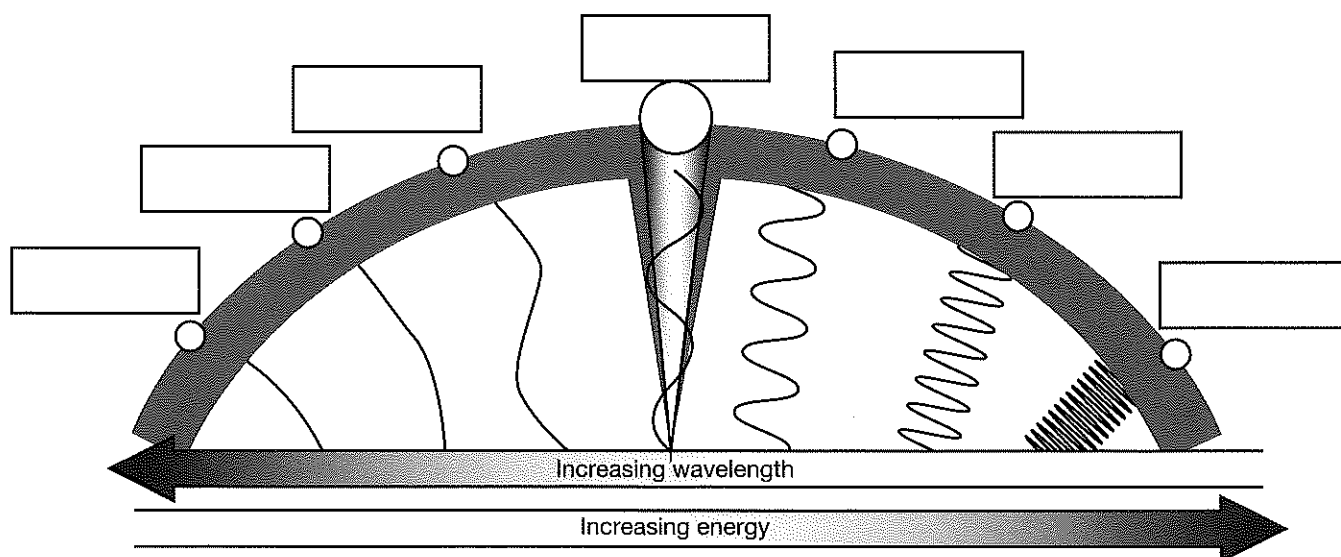
(h) Using the wave equation, $v = f\lambda$, calculate its frequency.

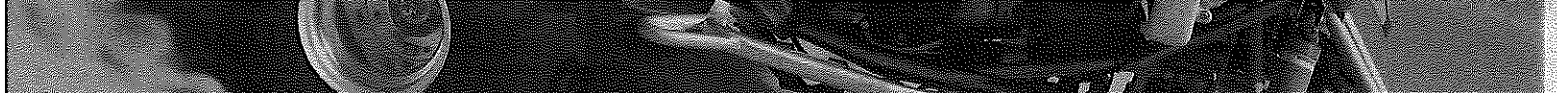
(i) If the frequency of this wave in air was doubled, what would happen to its speed? Explain your answer.

(j) If the frequency of this wave in air was doubled, what else would change? Explain your answer.



2.1.4.3 Identify the types of electromagnetic radiation labelled on the diagram. Write your answers in the spaces provided on the diagram.





- 2.1.4.4** Which electromagnetic radiation has the lowest frequency?
(A) Radio waves.
(B) Gamma rays.
(C) Infra-red waves.
(D) Ultraviolet waves.
- 2.1.4.5** If electromagnetic radiation is listed from gamma rays in order to TV waves, they are listed in order of:
(A) Decreasing amplitude.
(B) Decreasing frequency.
(C) Decreasing intensity.
(D) Decreasing wavelength.
- 2.1.4.6** What is the same for all electromagnetic radiation in a vacuum?
(A) Amplitude.
(B) Frequency.
(C) Speed.
(D) Wavelength.
- 2.1.4.7** Two types of electromagnetic waves are microwaves waves and X-rays. Which of these statements describes a property that is shared by microwaves and X-rays?
(A) Both have high frequencies.
(B) Both have long wavelengths.
(C) Both have high intensity.
(D) Both have the same speed in a vacuum.
- 2.1.4.8** Infra-red rays from the Sun travels through space:
(A) At half the speed of visible light.
(B) At the same speed as visible light.
(C) Slower than visible light.
(D) Faster than visible light.
- 2.1.4.9** How does visible EMR from the Sun differ from the non-visible EMR from the Sun?
(A) It has different amplitudes.
(B) It travels a different distance.
(C) It travels at a different speed.
(D) It has different frequencies.
- 2.1.4.10** Visible light has a higher frequency than:
(A) X-rays.
(B) Ultraviolet rays.
(C) Infra-red rays.
(D) Gamma rays.
- 2.1.4.11** What is the relationship between the wavelength (λ), and frequency (f), of an EMR?
(A) If λ increases, f decreases.
(B) If f increases, λ increases.
(C) If f remains constant, λ increases.
(D) If λ remains constant, f decreases.
- 2.1.4.12** Which statement is true regarding mechanical and electromagnetic waves?
(A) Both types of waves require a medium.
(B) Neither type requires a medium.
(C) A mechanical wave requires a medium, but an electromagnetic wave does not.
(D) EMR speed is affected by a medium, but the speed of a mechanical wave is not.
- 2.1.4.13** Which photons carry the highest energy?
(A) Red photons.
(B) Yellow photons.
(C) Blue photons.
(D) Violet photons.
- 2.1.4.14** The energy carried by electromagnetic waves depends on their:
(A) Amplitude.
(B) Frequency.
(C) Period.
(D) Wavelength.
- 2.1.4.15** Which of the following electromagnetic waves has the longest wavelength?
(A) Radio waves.
(B) Gamma rays.
(C) X-rays.
(D) Visible light.

2.1.5 The wave equation.

2.1.5.1 A wave has a wavelength of 0.25 m and a period of 4×10^{-4} s. What is its frequency and velocity?

2.1.5.2 330 m s⁻¹ soundwaves have a frequency of 600 Hz. What is their wavelength?

2.1.5.3 A tuning fork has a frequency 384 Hz. If the velocity of sound in air is 320 m s⁻¹, find the wavelength and period of the sound it produces.

2.1.5.4 A wave has a frequency of 2000 Hz and a wavelength of 0.2 cm. What is its velocity?

- (A) 4 m s⁻¹
- (B) 40 m s⁻¹
- (C) 400 m s⁻¹
- (D) 10 000 m s⁻¹

2.1.5.5 What is the wavelength of a 50 Hz periodic wave moving at 330 m s⁻¹?

- (A) 0.15 m
- (B) 6.6 m
- (C) 280 m
- (D) 16 500 m

2.1.5.6 If the frequency of a wave is 8.00 Hz and the wave speed is 48.00 m s⁻¹, what is the distance between successive wave crests on this wave?

- (A) 0.17 m
- (B) 6.0 m
- (C) 40 m
- (D) 384 m

2.1.5.7 A wave completes one vibration as it moves a distance of 1.2 metres at a speed of 15 m s⁻¹. What is the frequency of the wave?

- (A) 0.08 Hz
- (B) 12.5 Hz
- (C) 16.2 Hz
- (D) 18 Hz

2.1.5.8 330 m s⁻¹ soundwaves have a frequency of 125 Hz. What is their wavelength?

- (A) 6.4×10^{-5} m
- (B) 0.38 m
- (C) 2.64 m
- (D) 4125 m

2.1.5.9 What is the frequency of a wave that has a speed of 0.3 m s⁻¹ and a wavelength of 0.06 m?

- (A) 0.018 Hz
- (B) 0.05 Hz
- (C) 0.2 Hz
- (D) 5 Hz

2.1.5.10 What is the frequency of a wave with a period of 0.025 s?

- (A) 4 Hz
- (B) 40 Hz
- (C) 400 Hz
- (D) 4000 Hz

2.1.5.11 Find the speed of a wave with a wavelength of 2.5 cm and a frequency of 16 Hz.

- (A) 0.40 m s⁻¹
- (B) 0.86 m s⁻¹
- (C) 4.0 m s⁻¹
- (D) 40 m s⁻¹

2.1.5.12 A 350 Hz wave has a wavelength of 0.15 m. How long does it take to travel 200 m?

- (A) 0.26 s
- (B) 1.17 s
- (C) 3.81 s
- (D) 10.5 s

2.1.5.13 Consider a beam of red light where the photons have a wavelength of 750 nm. What will be their frequency?

- (A) 225 Hz
- (B) 400 000 Hz
- (C) 4×10^{11} Hz
- (D) 4×10^{14} Hz

2.1.5.14 A beam of blue photons has a frequency of 6×10^{14} Hz. What is their wavelength?

- (A) 5 nm
- (B) 50 nm
- (C) 500 nm
- (D) 5.0 m

2.2 Wave diagrams, period, amplitude, wavelength, frequency and velocity.

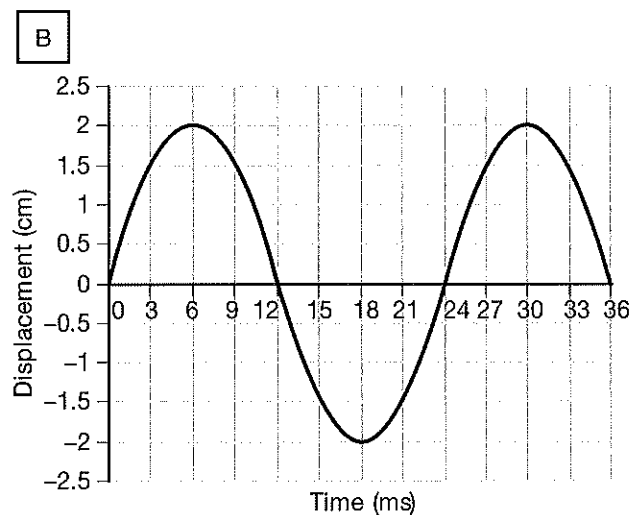
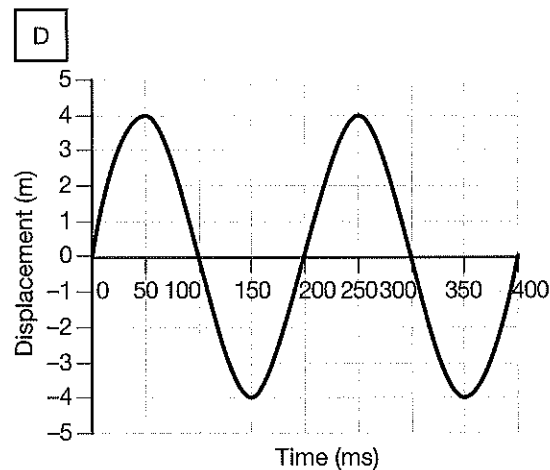
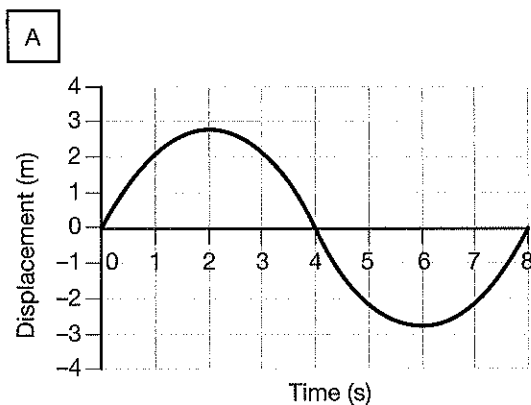
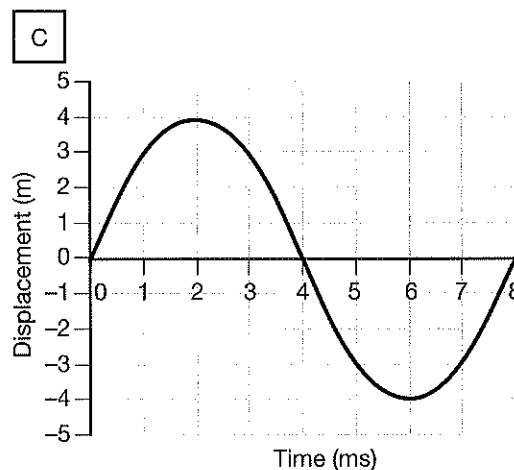
2.2.1 Analysing wave diagrams 1.

2.2.1.1 The graphs show the displacement of a particle in several waves, each of wavelength 1.25 m.

Determine, for each wave, the:

- (a) Period.
- (b) Frequency.
- (c) Wavelength.
- (d) Amplitude.
- (e) Speed.

Write your answers in the table provided.

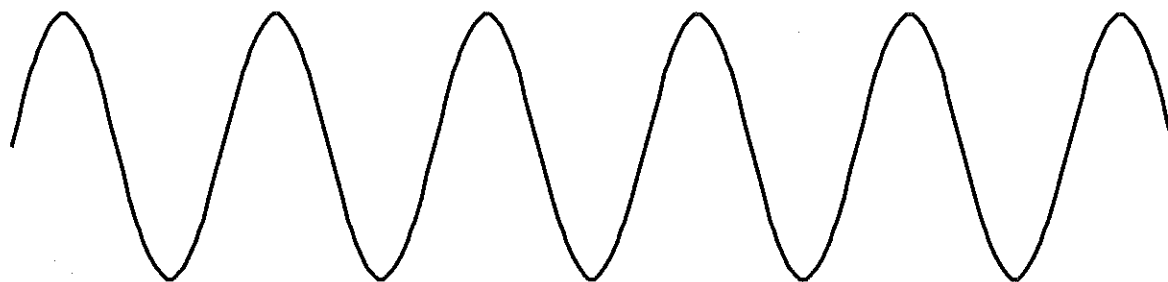


Wave	A	B	C	D
Period (s)				
Frequency (Hz)				
Wavelength (m)				
Amplitude (m)				
Velocity (m s^{-1})				

2.2.1.2

Below are shown several waves and the length of time each has been travelling. Analyse each to determine its wavelength, amplitude (A, B and C only), frequency, period and velocity. Each wave is shown actual size. Write your answers in the table.

A = 1.4 s



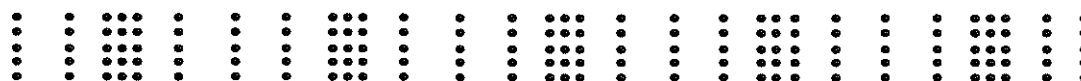
B = 2.5 s



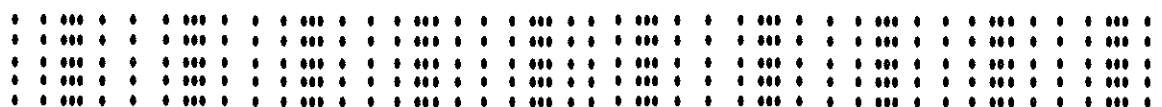
C = 0.025 s



D = 0.02 s



E = 4.3×10^{-4} s



F = 4×10^{-3} s

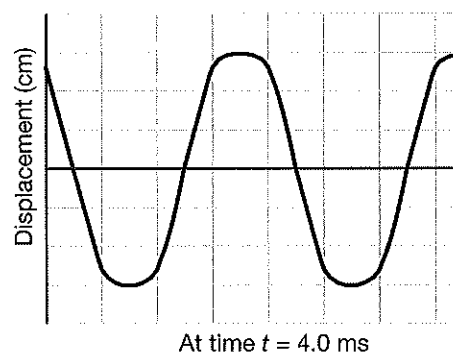
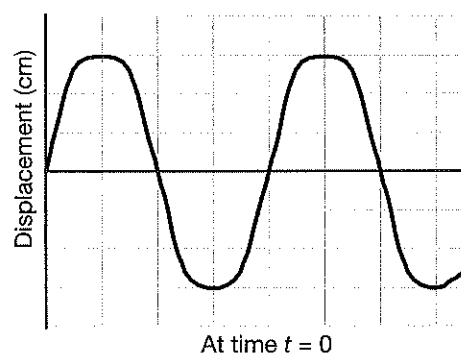


Wave	Wavelength (cm)	Amplitude (cm)	Frequency (Hz)	Period (s)	Velocity (m s ⁻¹)
A					
B					
C					
D		-			
E		-			
F		-			

2.2.2 Analysing wave diagrams 2.

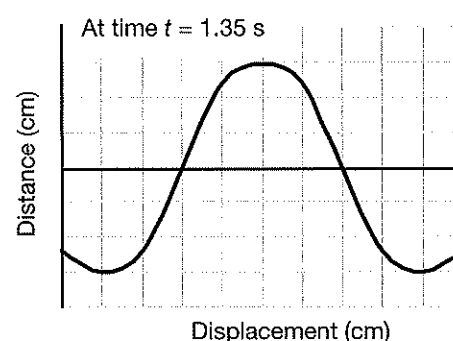
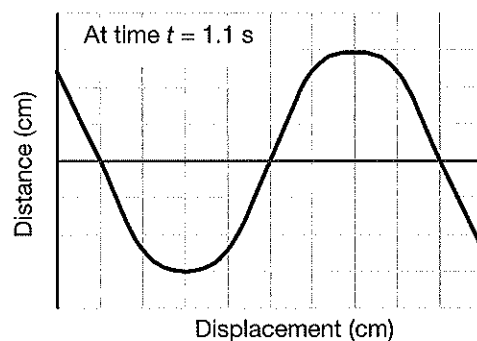
2.2.2.1 The graphs show two positions of a wave 4.0 ms apart.

Calculate the maximum period and the frequency of the wave.

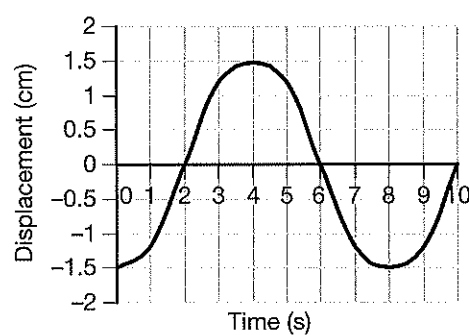
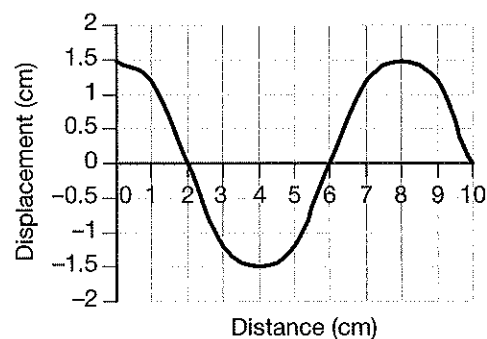


2.2.2.2 The graphs show two positions of a wave 0.25 s apart. Calculate the maximum period and the frequency of the wave.

Note that the x-axis divisions are different to those in the previous question.

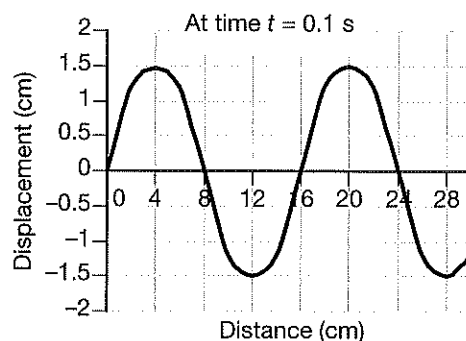
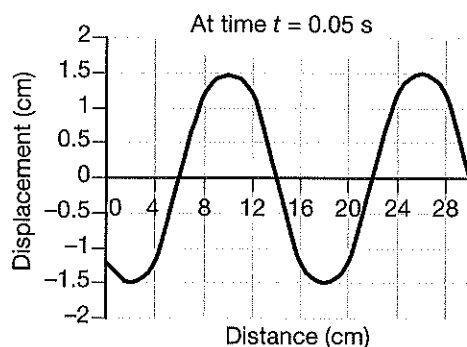


2.2.2.3 Analyse the graphs to determine the wavelength, period, frequency, amplitude and speed of the wave.



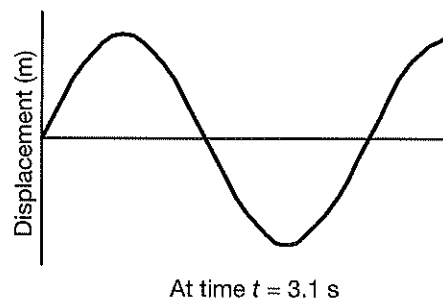
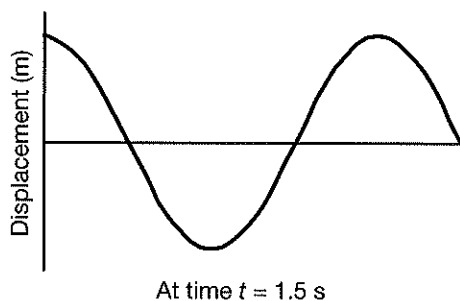
2.2.2.4 The graphs show two positions of a wave 0.05 s apart.

Calculate the wavelength, period, frequency, amplitude and speed of the wave.

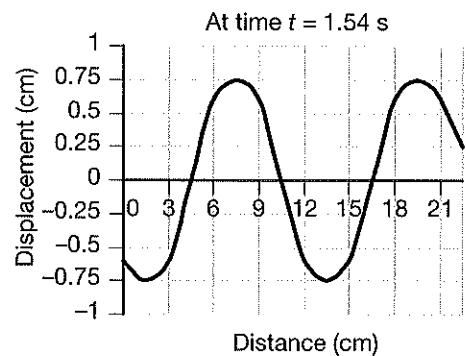
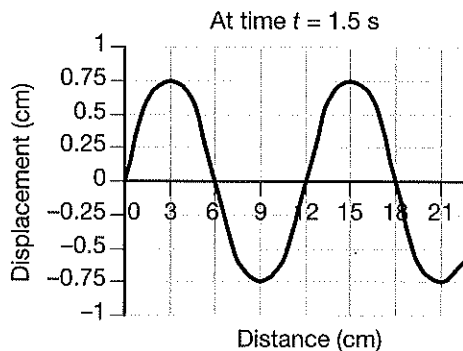


2.2.2.5 The graphs show the positions of the same wave 1.6 s apart.

Analyse them to calculate the maximum possible period for the wave.



2.2.2.6 The graphs show two positions of a wave 0.04 s apart. Calculate the wavelength, maximum period, frequency, amplitude and speed of the wave.



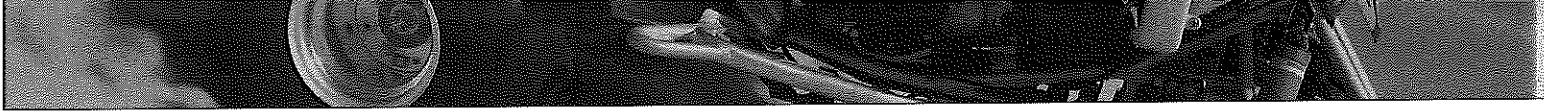


2.2.3 More about soundwaves.

2.2.3.1 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise ideas about soundwaves.

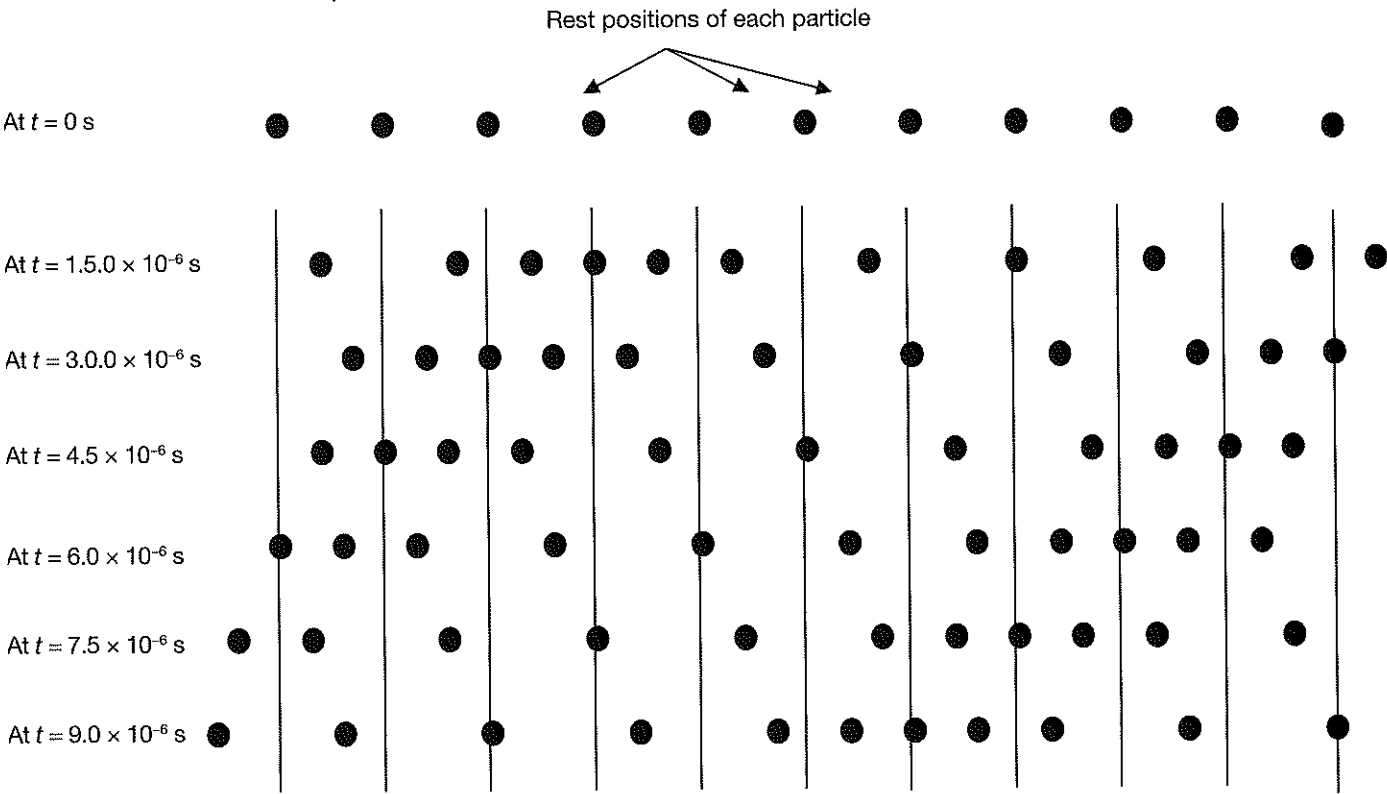
Completed sentences	Second halves (not in order)
(a) A soundwave is a means of transporting	as changes in pressure with respect to time.
(b) A soundwave is a pressure wave. It can be considered	because solids are more dense.
(c) If the frequency of a sound in a particular	the wavelength but not alter the speed of the wave.
(d) In general, soundwaves travel fastest in	the properties of the medium through which it moves.
(e) Soundwaves travel fastest in solids (compared to liquids and gases)	fastest in the most dense materials.
(f) A highly elastic material has a strong tendency to return	energy without transporting matter.
(g) A rigid material such as steel has a high elasticity and therefore sound tends to	to its original shape if stressed, stretched, plucked or somehow disturbed.
(h) The speed of a soundwave depends only on	through air at the same speed.
(i) If all other factors are equal, a soundwave will travel	solids and slowest in gases.
(j) Both low and high pitched sounds will travel	air at the same speed.
(k) Doubling the frequency of the source of a soundwave will halve	move through it at higher speeds that, say, through air.
(l) Both low and high pitched sounds will travel through	medium is increased then its wavelength will decrease.

- 2.2.3.2** What term do we use to describe the 'loudness' or 'softness' of a sound?
- (A) Amplitude.
 - (B) Frequency.
 - (C) Intensity.
 - (D) Pitch.
- 2.2.3.3** If the velocity of a soundwave is 330 m s^{-1} and its frequency is 250 Hz , what is its period?
- (A) $3.03 \times 10^{-3} \text{ s}$
 - (B) $4.0 \times 10^{-3} \text{ s}$
 - (C) 0.76 s
 - (D) 1.32 s
- 2.2.3.4** Which one of the following factors determines the pitch of a sound?
- (A) The amplitude.
 - (B) The distance of the wave from its source.
 - (C) The frequency.
 - (D) Its wavelength.
- 2.2.3.5** What does the speed of a longitudinal wave through a medium depend upon?
- (A) Its amplitude.
 - (B) Its frequency.
 - (C) The medium it travels through.
 - (D) Its wavelength.
- 2.2.3.6** What happens to the particles in the air when a soundwave travels through it?
- (A) They vibrate but not in any fixed direction.
 - (B) They vibrate along the direction of wave propagation.
 - (C) They vibrate perpendicular to the direction of wave propagation.
 - (D) They remain in position relative to each other.
- 2.2.3.7** Which statement is true? Soundwaves will travel fastest through:
- (A) Gases.
 - (B) Liquids.
 - (C) Solids.
 - (D) A vacuum.
- 2.2.3.8** A wave with a frequency of $X \text{ Hz}$ through a medium with a speed of $Y \text{ m s}^{-1}$. The wavelength of the wave could be expressed as:
- (A) $\frac{Y}{X}$
 - (B) $\frac{X}{Y}$
 - (C) XY
 - (D) $X + Y$
- 2.2.3.9** Sound travels slower in air at:
- (A) Lower temperatures because molecules move faster and collide less often.
 - (B) Higher temperatures because molecules are closer together and collide less often.
 - (C) Lower temperatures because molecules are closer together and collide less often.
 - (D) Lower temperatures because molecules move slower and collide less often.
- 2.2.3.10** A 500 Hz sound travels through air at 360 m s^{-1} . What is the distance between an adjacent compression and rarefaction in this wave?
- (A) 0.36 m
 - (B) 0.72 m
 - (C) 1.39 m
 - (D) 0.69 m
- 2.2.3.11** What is the distance between a rarefaction and the next compression in a soundwave?
- (A) $\frac{\lambda}{8}$
 - (B) $\frac{\lambda}{4}$
 - (C) $\frac{\lambda}{2}$
 - (D) 2λ
- 2.2.3.12** A soundwave has a wavelength of 6.0 m . What is the distance between the centre of a compression and the centre of the next rarefaction?
- (A) 0.375 m
 - (B) 0.75 m
 - (C) 1.5 m
 - (D) 3.0 m



2.2.3.13 The diagrams show the rest positions of the particles in a medium and their positions at several instances of time as a wave motion passes through the medium.

The rest positions on the particles are shown as vertical lines in the diagrams. The distance between successive rest positions is $2.5 \times 10^{-8} \text{ m}$.



Analyse the diagrams to find the following.

- (a) The wavelength.
-
- (b) The period.
-
- (c) The frequency.
-
- (d) The amplitude.
-
- (e) The speed of the wave.
-

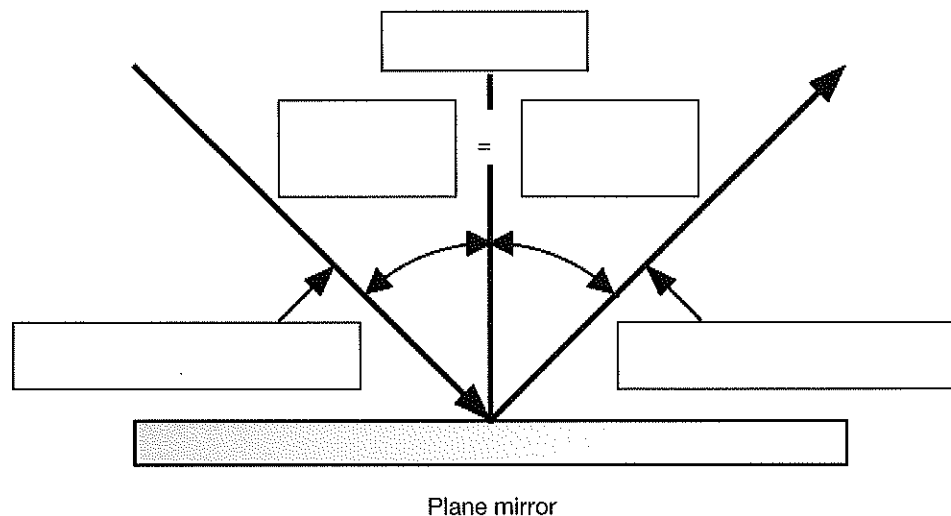
2.3 The ray model of light.

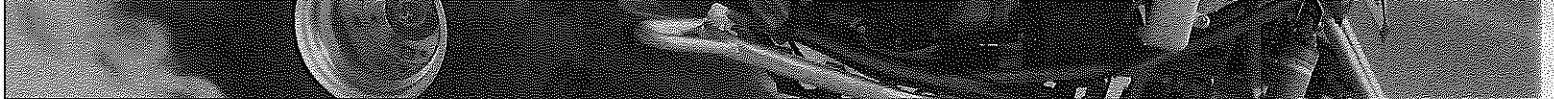
2.3.1 Reflection from plane surfaces.

2.3.1.1 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise ideas about reflection.

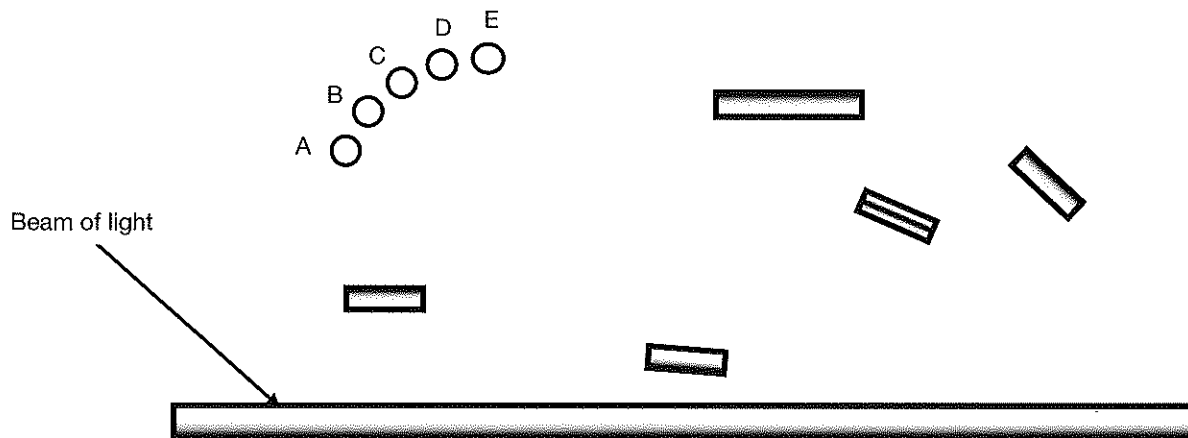
Completed sentences	Second halves (not in order)
(a) Reflection refers to the way waves 'bounce back'	of incidence is equal to the angle of reflection.
(b) We see things because reflected	into the surface and the normal to the surface.
(c) The law of reflection states that the angle	the ray reflected away the surface and the normal to the surface.
(d) The angle of incidence is the angle between the ray coming	when they hit a surface.
(e) The ray coming into a surface	is known as the reflected ray.
(f) The angle of reflection is the angle between	is known as the incident ray.
(g) The ray reflected away from a surface	light from those things travels to our eyes.

2.3.1.2 Use the information from your completed sentences to label the diagram below.

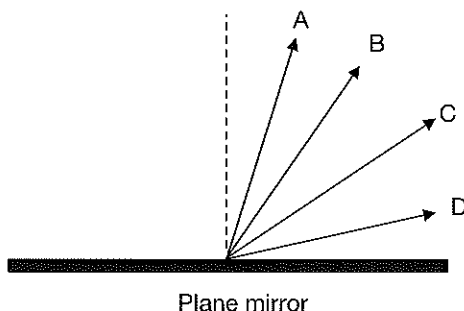




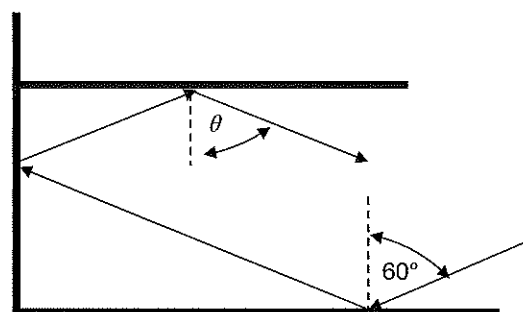
- 2.3.1.3** Complete the following diagrams by drawing in any missing normals, incident rays or reflected rays to determine which circle the reflected ray eventually hits.



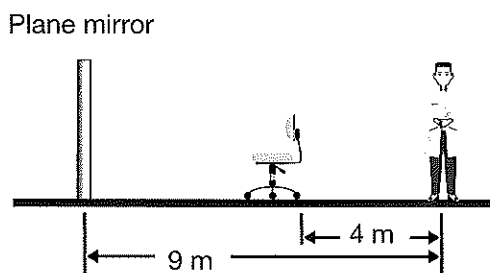
- 2.3.1.4** On the diagram below, draw in the incident ray which would give B as the reflected ray.



- 2.3.1.6** Find the angle θ in the following diagram.



- 2.3.1.5** A person is standing 9 m from a plane mirror. The chair is 4 m in front of the person. What is the distance between the person and the image he observes of the chair?



- (A) 4 m (B) 13 m
(C) 14 m (D) 18 m

- 2.3.1.7**

- (a) A window is covered with a light reflecting plastic film to give privacy to the people in the room during the daytime, but at the same time allowing them uninterrupted vision through the window to the outside. Explain why this works.

.....

.....

.....

.....

- (b) Will the film give the same privacy during the night? Explain your answer.

.....

.....

.....

.....

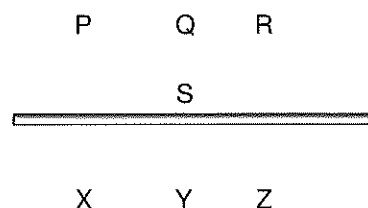
2.3.1.8 A ray of light hits a mirror at an angle of incidence of 46° . What is the angle between the incident and reflected rays?

- (A) 23° (B) 46° (C) 88° (D) 92°

2.3.1.9 Persons X, Y and Z are standing near each other the same distance in front of a plane mirror as shown in the diagram.

Which statement is correct?

- (A) X will see Z's reflection at P.
 (B) Z will see X's reflection at R.
 (C) Y sees X's reflection at S.
 (D) X and Z see each other at Q.



2.3.1.10 Which states the law of reflection?

- (A) Angle $i = \text{angle } r$ (B) Angle $i > \text{angle } r$ (C) Angle $i < \text{angle } r$ (D) Angle $i \neq \text{angle } r$

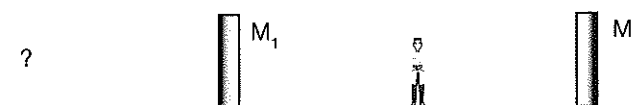
2.3.1.11 A boy is 8 m in front of and running towards the front mirrors at the front of the taekwondo dojang at 2.5 m s^{-1} . Which statement about the boy and his image in the mirrors is correct?

- (A) They approach each other at 2.5 m s^{-1} .
 (B) They approach each other at 5.0 m s^{-1} .
 (C) They approach each other at 10.5 m s^{-1} .
 (D) The boy approaches the mirror at 2.5 m s^{-1} but the relative position of his image is unchanged.

2.3.1.12 You stand midway between two mirrors which are 10 m apart looking at M_1 .

If you were to look at your image from M_1 in mirror M_2 behind you, how far would it be away from you?

- (A) 5 m (B) 10 m
 (C) 20 m (D) 30 m



2.3.1.13 Which of the following best describes the image formed by a plane mirror?

- (A) Virtual, inverted and enlarged. (B) Virtual, upright and the same size as object.
 (C) Real, inverted and reduced. (D) Real, upright and the same size as the object.

2.3.1.14 A man sits in a barber's chair, looking into a plane mirror, which is 1.5 m away from him. He looks at the image of a poster which is on the wall behind him, 3.0 m from the chair.

How far from him is the image of the poster?

- (A) 3.5 m (B) 4.5 m (C) 6.0 m (D) 9.0 m

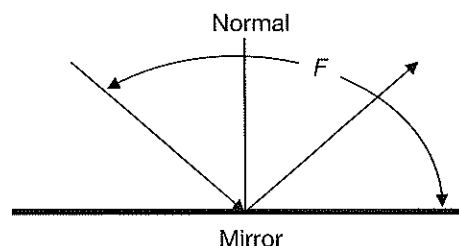
2.3.1.15 Why can't you see your reflection in a wooden fence?

- (A) Light does not reflect from a rough surface.
 (B) The light is absorbed rather than being reflected by the fence.
 (C) Light rays are reflected in many directions from the uneven surface.
 (D) Light rays are reflected in only one direction from the rough surface.

2.3.1.16 Consider the following diagram.

If angle F is 130° , what is the angle of reflection?

- (A) 30° (B) 40°
 (C) 50° (D) 60°



2.3.2 Reflection in a plane mirror – a practical.

2.3.2.1 Follow the instructions to complete the diagram below. The completed diagram will show how an image of an object is formed in a plane mirror.

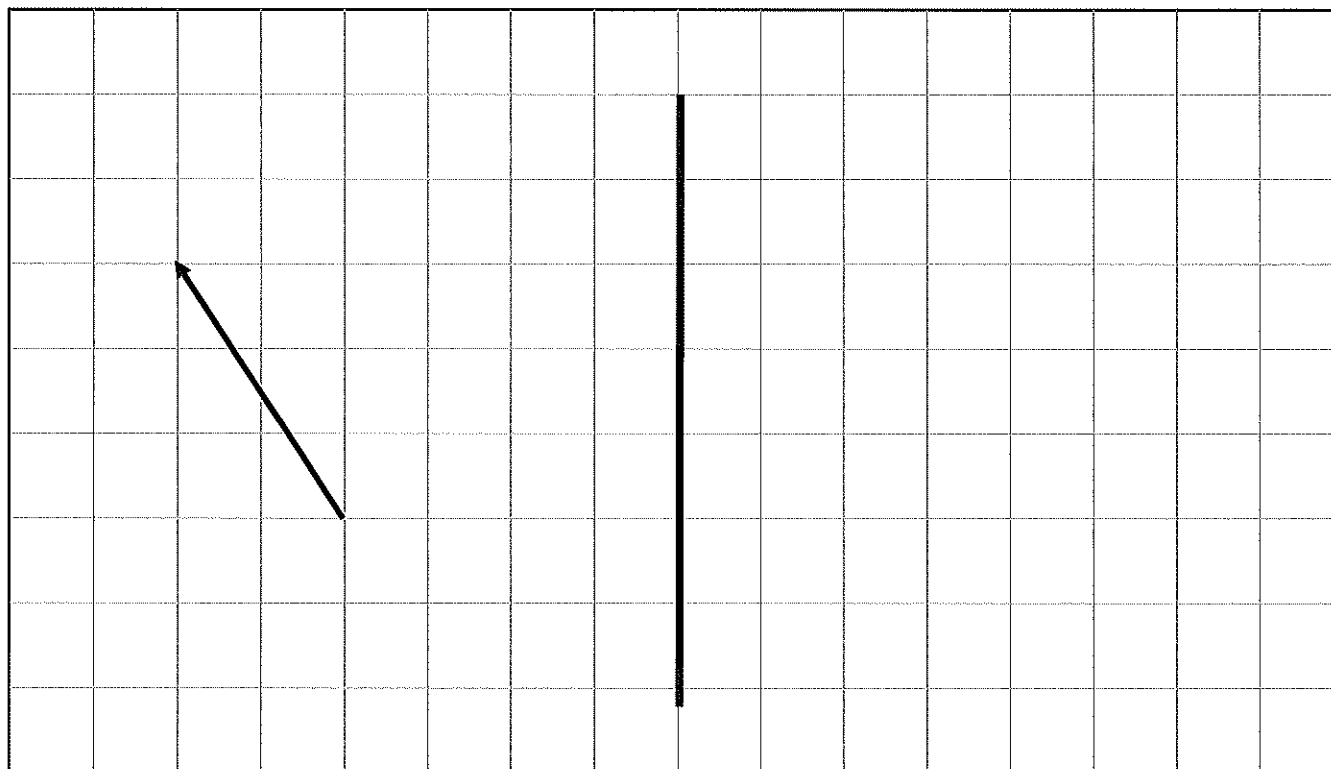
Instructions A – top of the object

1. Draw a light ray (not horizontal) from the top of the *object* to *anywhere* on the mirror.
2. Draw in a normal to the surface of the mirror where this ray is incident on the mirror.
3. Measure the angle of incidence and use this to draw in a red reflected ray.
4. *Extend* the reflected ray as a dotted line *behind* the mirror.
5. Repeat steps 1 to 4 with a second ray of light from the top of the object to a different point on the mirror.
6. The position behind the mirror where the two backwards extension lines meet marks the position of the top of the image. Draw the arrowhead here.

Instructions B – bottom of the object

7. Repeat steps 1 to 6 for two lines drawn from the bottom of the object.
8. Where the two backwards extension lines meet is the position of the bottom of the image.
9. Draw in the bottom of the image.
10. Connect the top and bottom to complete the image.
11. Measure the height of the image and compare this with the height of the object.
12. Measure the distance of the object in front of the mirror, and that of the image behind the mirror.
13. Use your measurements to complete the statements in the text box below.

- (a) An image formed in a plane mirror will be as the object is in front of the mirror.
- (b) The size of the image formed by a plane mirror will be as that of the image it forms.



2.3.3 Reflection from curved surfaces.

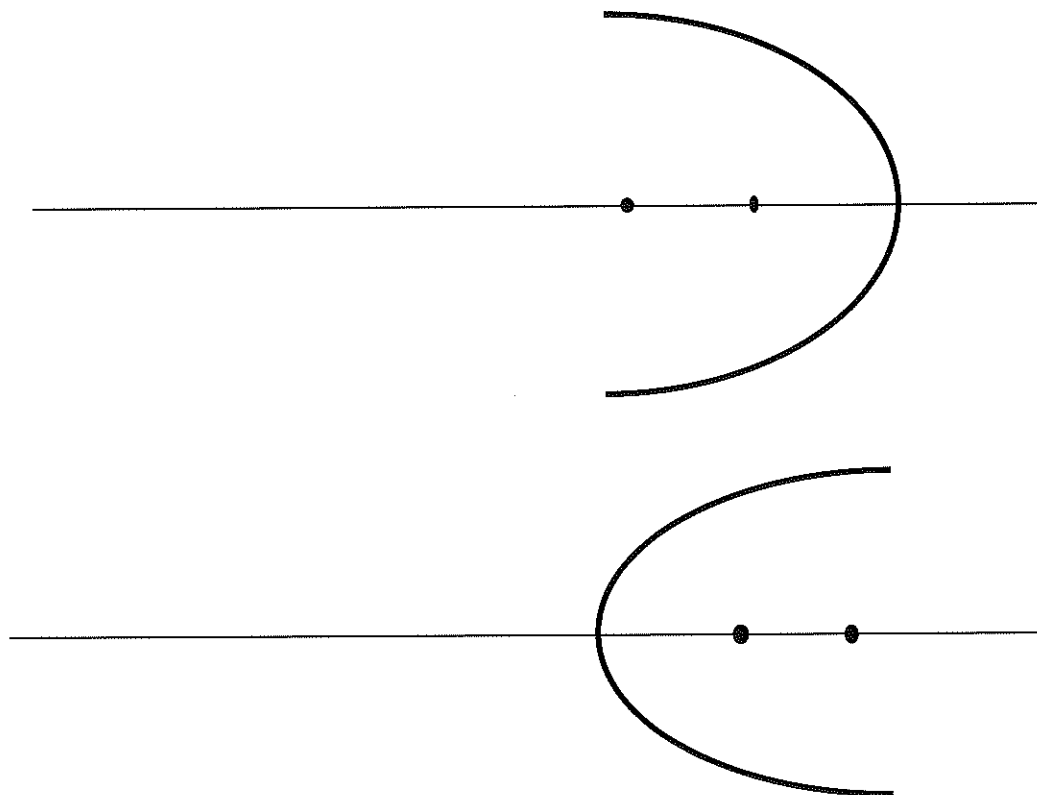
2.3.3.1 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise ideas about reflection from curved surfaces.

Completed sentences	Second halves (not in order)
(a) Surfaces which curve inwards (like a cave)	passes through the focus of the curve.
(b) Surfaces which curve outwards	reflected straight back (it hits the curved surface at 90°).
(c) Convex surfaces	are known as concave surfaces.
(d) Converging surfaces cause the rays to reflect and move closer	the focus of the curve.
(e) Convex surfaces reflect rays so that their backwards extension	axis through which reflected rays pass.
(f) Convex surfaces are called diverging surfaces because after	are converging surfaces.
(g) When electromagnetic rays hit a concave surface they reflect so that they pass through	through the centre of the lens.
(h) When electromagnetic rays hit a convex surface they reflect so that their extension	reflection the rays travel further apart.
(i) The ray travelling along the principal axis is	together until they pass through the focus.
(j) The principal axis is the (imaginary) line drawn	centre of curvature of the surface.
(k) The focus of a curved surface is the point on the principal	are called convex surfaces.
(l) The focus of a curved surface is midway between the surface and the	backwards passes through the focus of the curve.



2.3.3.2

- (a) On each of the following diagrams, label, where relevant: concave surface, convex surface, centre of curvature, focus, principal axis.
- (b) On each diagram draw in five incident rays, one along the principal axis, and their reflections.



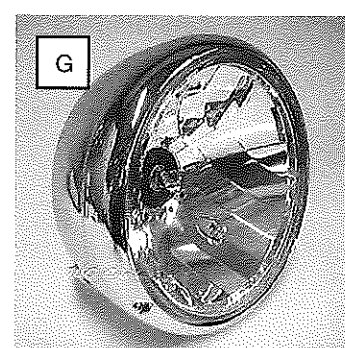
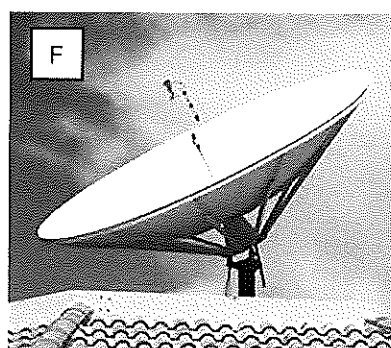
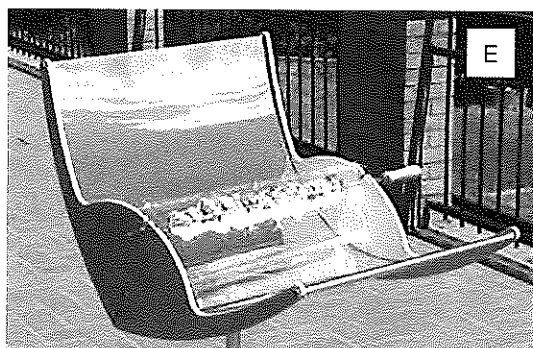
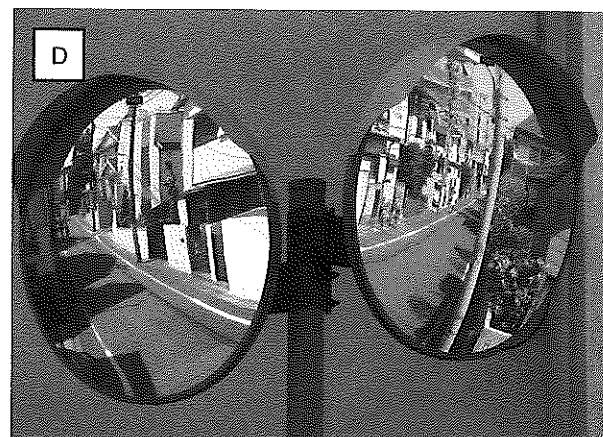
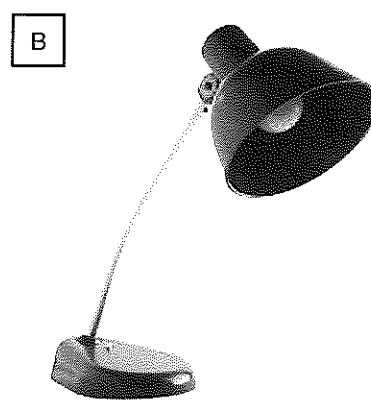
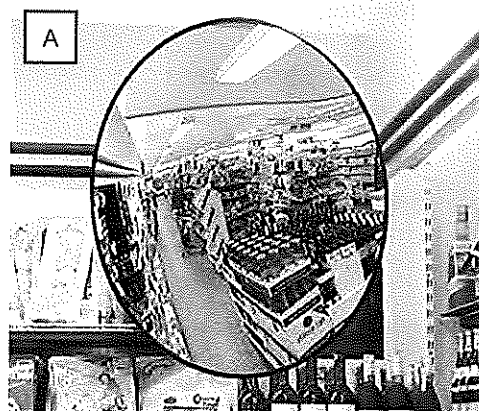
2.3.3.3

- (a) State three uses for convex reflective surfaces and explain the property of the surface that is essential for each of these uses.
-
-
-
- (b) State three uses for concave reflective surfaces and explain the property of the surface that is essential for each of these uses.
-
-
-
- (c) State three uses for plane reflective surfaces and explain the property of the surface that is essential for each of these uses.
-
-
-

2.3.3.4 The pictures show various uses for reflective surfaces.

For each, identify the type of surface and why it is used for the purpose shown in the picture.

Write your answers in the table provided.

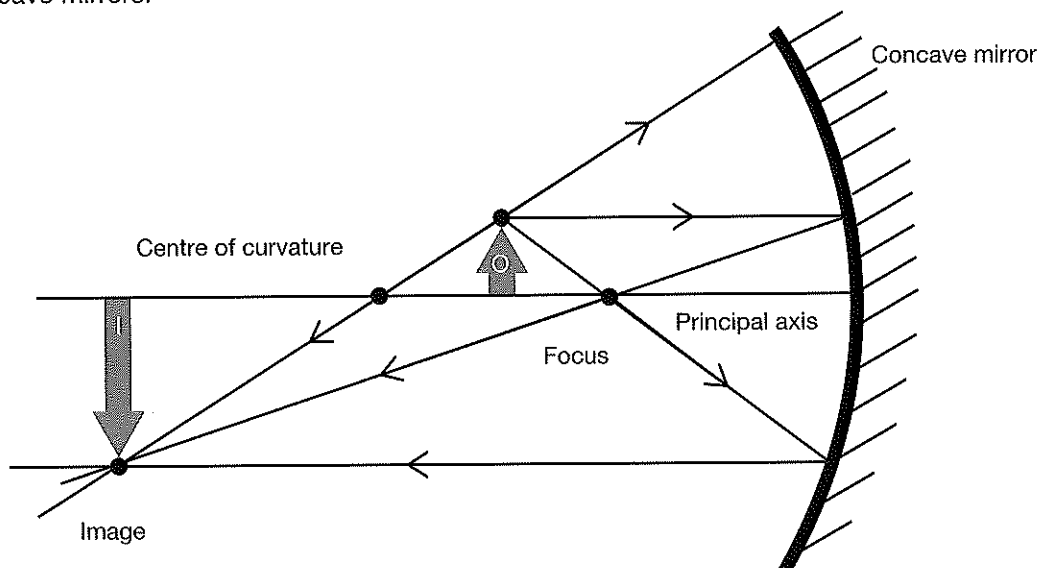


	Concave or convex reflector	Reason for use in this application
A		
B		
C		
D		
E		
F		
G		

2.3.4 Images formed by concave mirrors.

2.3.4.1 The diagram shows the formation of an image by a concave mirror, and the ray diagram we draw to find the position and nature of that image.

In doing this, the diagram represents the three main ideas we implement when drawing ray diagrams for concave mirrors.



Examine the diagram, then complete the three sentences to summarise the three ideas we use when drawing ray diagrams for concave mirrors.

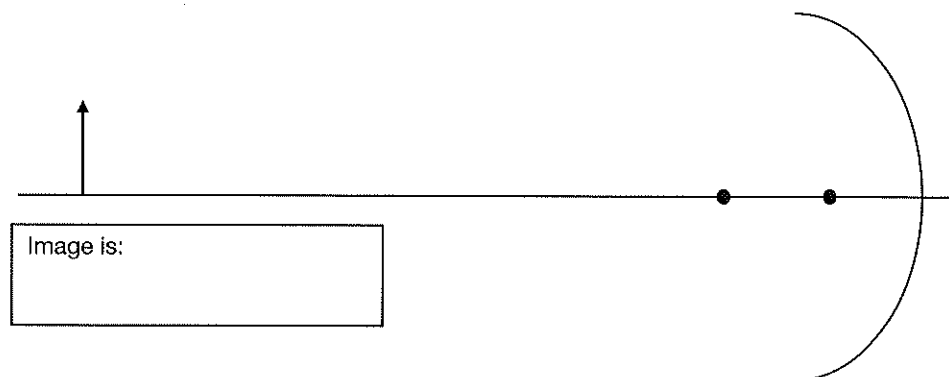
- A ray from the object travelling parallel to the principal axis will reflect
- A ray from the object that passes through the centre of curvature of a mirror is reflected
- Any ray from the object passing through the focus will be reflected

2.3.4.2 Complete the information in the table to describe the possible nature of images formed in concave mirrors.

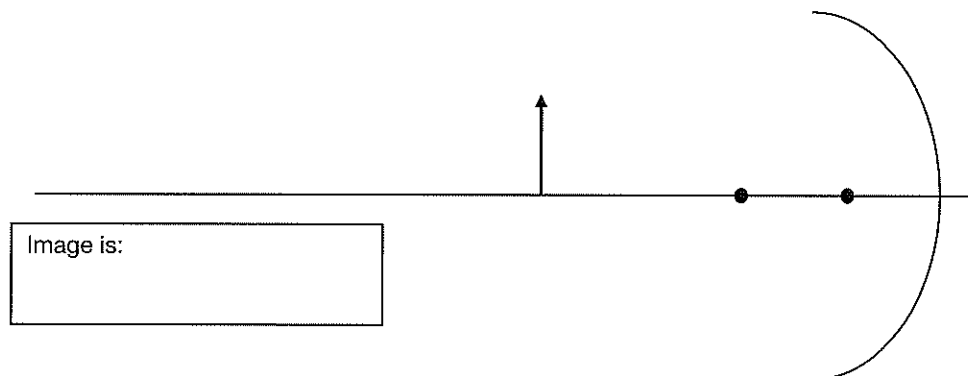
	Image is	Description of this nature
(a)	Real if	
(b)	Virtual if	
(c)	Upright if	
(d)	Inverted if	
(e)	Diminished if	
(f)	Diminished if	
(g)	Same size if	

2.3.4.3 Draw ray diagrams to determine the nature of the images formed by the concave mirror for each of the following positions of an image. Describe the nature of each image.

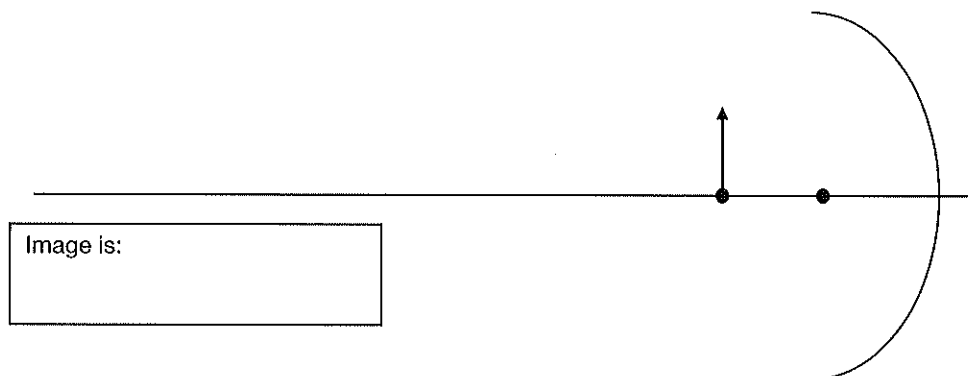
- (a) Object at a significant distance from mirror.

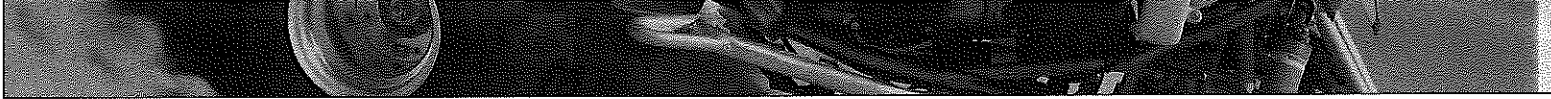


- (b) Object beyond C but not significantly distant.

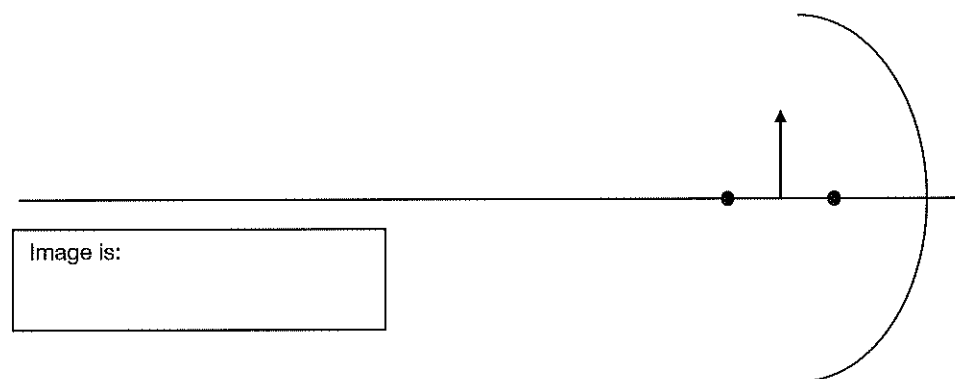


- (c) Object at C.

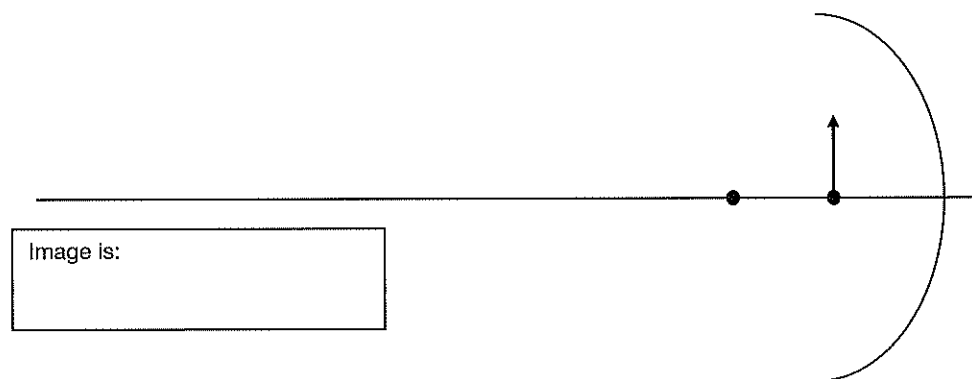




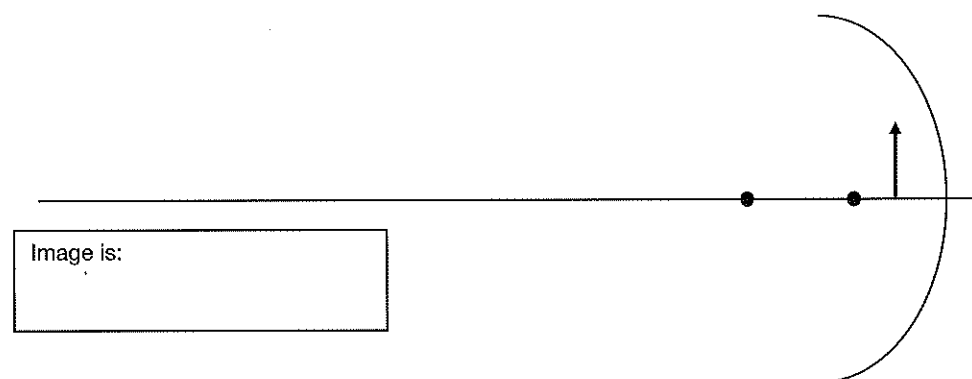
(d) Object between C and F.



(e) Object at F.



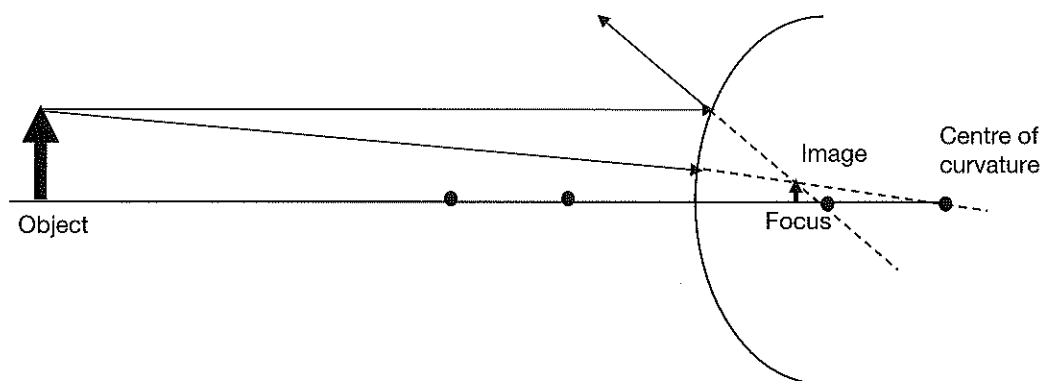
(f) Object between F and mirror.



2.3.5 Images formed by convex mirrors.

2.3.5.1 The diagram shows the formation of an image by a convex mirror, and the ray diagram we draw to find the position and nature of that image.

In doing this, the diagram represents the two main ideas we implement when drawing ray diagrams for convex mirrors.

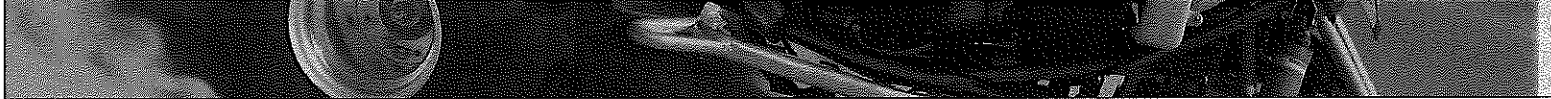


Examine the diagram, then complete the three sentences to summarise the three ideas we use when drawing ray diagrams for convex mirrors.

- (a) A ray from the object travelling parallel to the principal axis will reflect
- (b) A ray from the object that passes through the centre of curvature of a mirror is reflected
- (c) Any ray from the object passing through the focus will be reflected

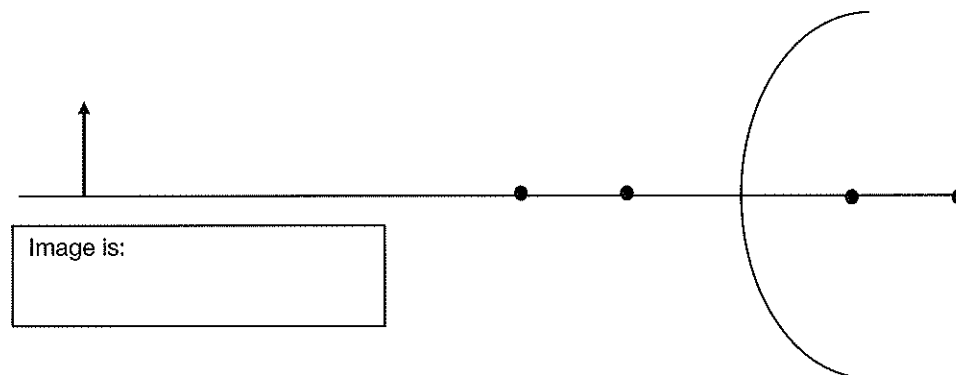
2.3.5.2 Complete the information in the table to describe the possible natures of images formed in convex mirrors.

	Image is	Description of this nature
(a)	Real if	
(b)	Virtual if	
(c)	Upright if	
(d)	Inverted if	
(e)	Diminished if	
(f)	Diminishes if	
(g)	Same size if	

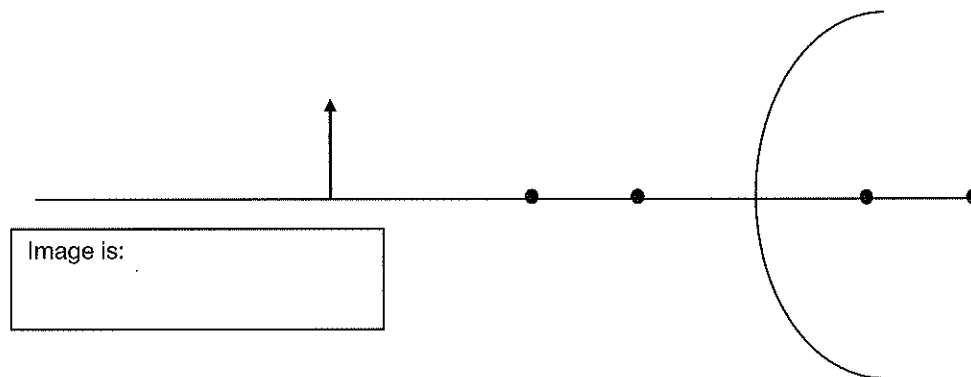


2.3.5.3 Draw ray diagrams to determine the nature of the images formed by the convex mirror for each of the following positions of an image. Describe the nature of each image.

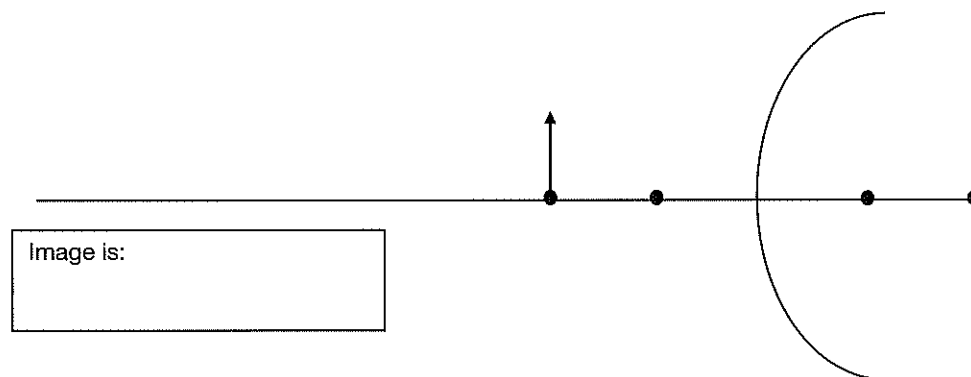
- (a) Object at a significant distance from mirror.



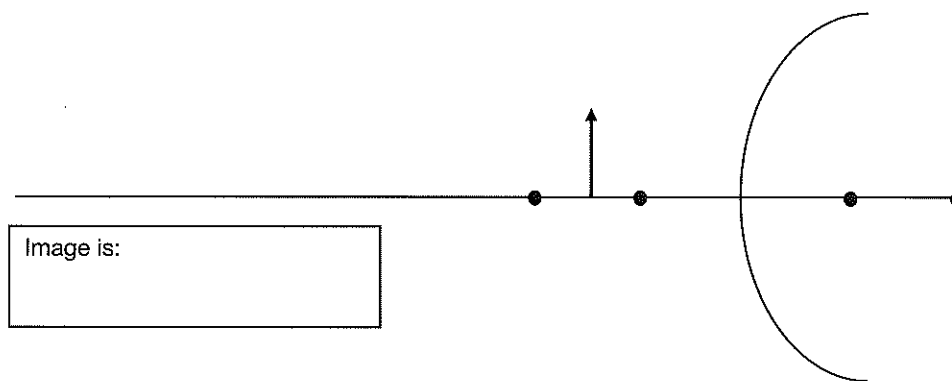
- (b) Object beyond C but not significantly distant (equivalent distance).



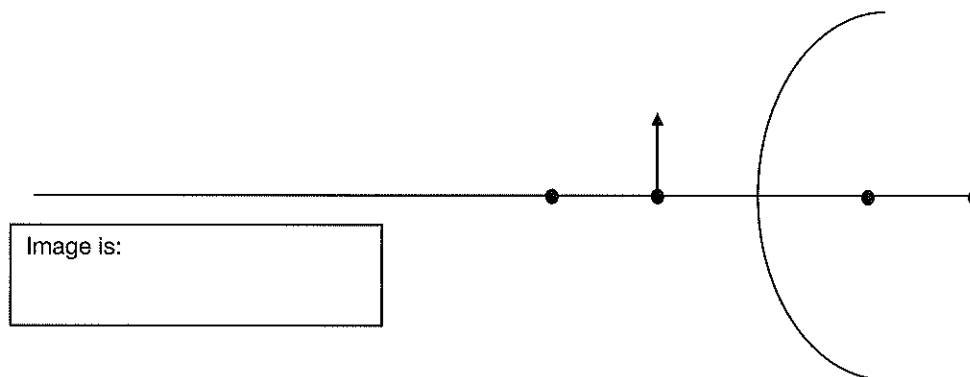
- (c) Object at C (equivalent distance).



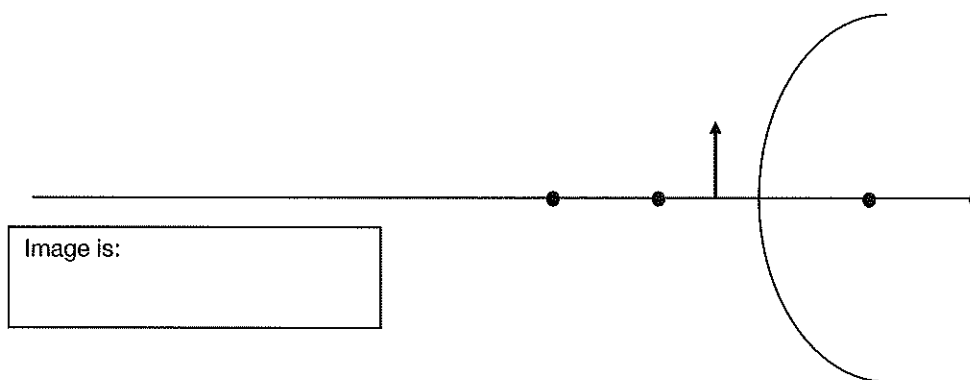
- (d) Object between C and F (equivalent distance).



- (e) Object at F (equivalent distance).



- (f) Object between F and mirror (equivalent distance).



2.3.6 Applications of reflection of sound.

2.3.6.1 A person is standing between two vertical cliffs. He is 750 m away from the nearest cliff and shouts.

(a) If the velocity of sound in the air is 330 m s^{-1} , how long will it be before he hears his echo?

(b) He hears the second echo from the cliff behind him 4.2 s after he hears the first echo. How far apart are the cliffs?

2.3.6.2 A ship on the surface of water sends a signal and receives it back after 8.5 seconds from a submarine under the water. If the speed of sound in the water is 1450 m s^{-1} , calculate the distance of the submarine from the ship.

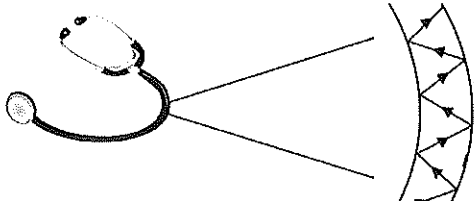
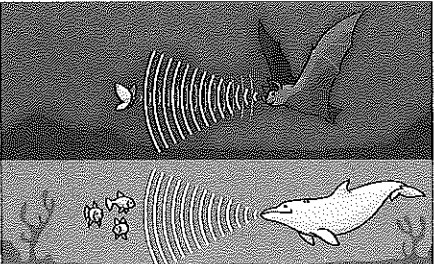
2.3.6.3 A man fires a gun and hears its echo from a cliff in front of him after 4.50 seconds. The man then moves 250 m closer to the cliff and fires his gun again. This time he hears the echo after 2.98 seconds. Calculate

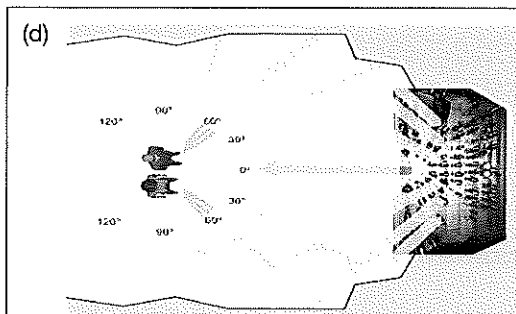
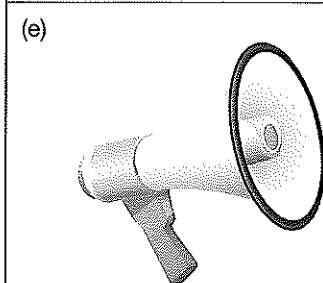
(a) The speed of sound in the air

(b) How far he was from the cliff in front of him when he fired the first shot.

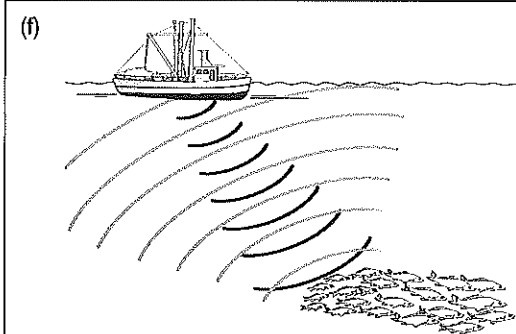
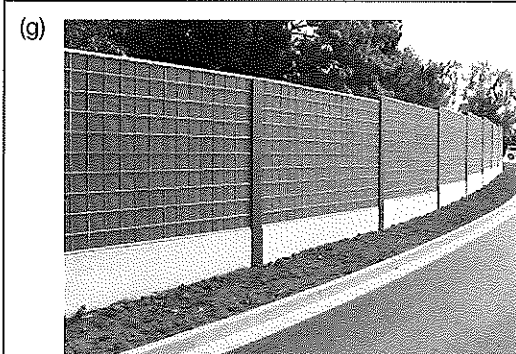
2.3.6.4 An ultrasonic wave is sent from a ship towards the bottom of the sea. It is found that the time interval between the sending and receiving of the wave is 3.4 s. If the velocity of sound in the sea water is 1400 m s^{-1} , what is the depth of the sea?

2.3.6.5 Each of the illustrations shows various devices which use the reflection of soundwaves in their operation. Identify each device and briefly describe how the reflection of sound is used in its operation.

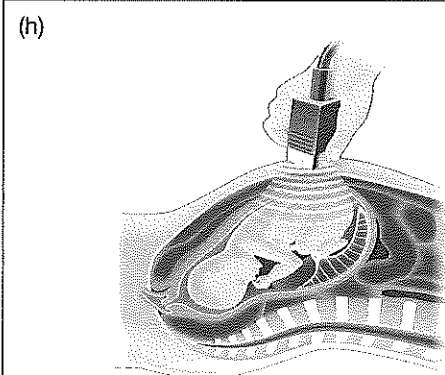
(a) 	
(b) 	
(c) 	

[illegible]

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#####
#####
#####
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[illegible]

.....



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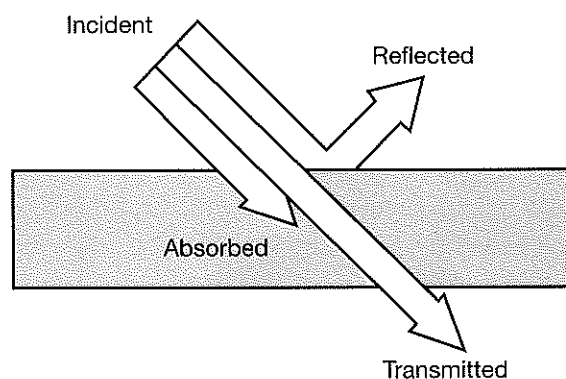
2.4 The wave model of light.

2.4.1 The wave theory of light.

2.4.1.1

- (a) Who, and when, in attempting to explain colour, was the first person to propose that light had a wave nature?
.....
- (b) Who, and when, proposed the wave theory of light as we understand it?
.....
- (c) This scientist was also the first to propose why light rays which are incident on glass at an angle other than 90° bent as they travelled through the glass. What was his explanation?
.....
- (d) What name do we give now to the property of light (in fact all electromagnetic radiation and other types of waves also) whereby its speed changes when it enters a different medium?
.....
- (e) Who, 100 years later made observations on the interference of light which can only be explained if light does have a wave nature?
.....
- (f) Outline the proposals made by scientists about the interference of light which caused them to support the wave nature of light.
.....
.....
- (g) Who, and why, was the greatest opponent to scientists proposing a theory of light in opposition to the particle theory of light?
.....
.....

- 2.4.1.2** Explain how reflection and transmission of light through a glass slab, as illustrated in the diagram was taken as evidence that light has a wave nature and is not made up of particles.
-
-
-
-
-
-



2.4.1.3

- (a) What is diffraction?
- (b) How does the wave theory of light explain diffraction?
- (c) How does the particle theory of light explain diffraction?

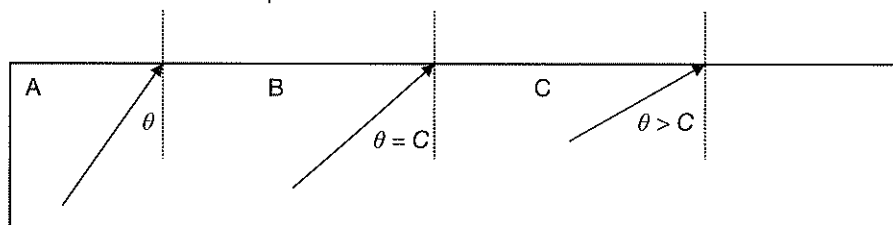
2.4.1.4

- (a) What is meant by the dispersion of light?
- (b) Explain how the wave theory of light would explain dispersion.

2.4.2 Total internal reflection.

2.4.2.1

- (a) What is total internal reflection?
- (b) What are the two conditions required for total internal reflection to occur?
- (c) What is the critical angle?
- (d) Complete the diagram below to show what happens to each incident ray when it hits the boundary. Accurate measurements are not required.



- (e) What is the most common use we have for total internal reflection?
- (f) Using Snell's law, derive an expression for the relationship between the critical angle of a substance and the relative refractive index of the boundary involved.

2.4.3 Refraction 1.

2.4.3.1

(a) What is refraction?

(b) Complete these sentences.

- (i) During refraction the velocity of a wave
- (ii) During refraction the frequency of a wave
- (iii) During refraction the wavelength of a wave

2.4.3.2 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise ideas about the refraction that occurs when electromagnetic waves travel from one medium into another. Some may be used more than once.

Completed sentences	Second halves (not in order)
(a) Electromagnetic waves slow down as they pass from	smaller in denser media.
(b) Electromagnetic waves speed up as they pass from	stays the same.
(c) The wavelength of electromagnetic waves is	decreases due to some reflection from the boundary.
(d) The wavelength of electromagnetic waves is	more dense to less dense media.
(e) The frequency of an electromagnetic wave passing from a less dense to a more dense medium	larger in less dense media.
(f) The frequency of an electromagnetic wave passing from a more dense to a less dense medium	less dense to more dense media.
(g) The intensity of an electromagnetic radiation passing from a less dense to a more dense medium	
(h) The intensity of an electromagnetic radiation passing from a more dense to a less dense medium	

2.4.3.3 What causes light waves to refract?

- (A) The speed of light is reduced to zero.
- (B) The speed of light remains constant.
- (C) The light is repelled by some materials.
- (D) The light changes from one medium to another.

2.4.3.4 A beam of light travels from air into a different medium. Which of the following will refract the light the least?

- (A) Diamond. (B) Glass. (C) Plastic. (D) Water.

2.4.3.5 In which situation would light speed up?

- (A) Passing from glass to air.
- (B) Passing from water to plastic.
- (C) Passing from air to plastic.
- (D) Passing from a vacuum to air.

2.4.3.6 Which of the following statements about the speed of light is true?

- (A) It is fastest in diamond.
- (B) It is slower in glass than in air.
- (C) It is faster in glass than in water.
- (D) It is the same speed in glass and diamond.

2.4.3.7 An electromagnetic wave speeds up as it passes from medium X into Y. Which of the following statements is true about X and Y?

- (A) X could be glass and Y air.
- (B) X could be water and Y plastic.
- (C) X could be air and Y plastic.
- (D) X could be plastic and Y glass.

2.4.3.8 Which statement about the speed of an electromagnetic wave is true?

- (A) It is not affected by changes in media.
- (B) It is slowest in a vacuum.
- (C) Its speed in a vacuum is slightly more than its speed in air.
- (D) It slows down when it goes into a less dense medium.

2.4.3.9 As a wave is refracted, which properties of the wave change?

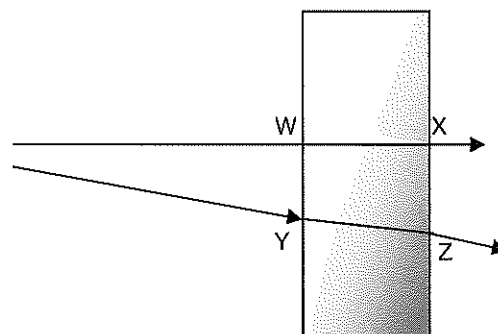
- (A) Only its velocity.
- (B) Its velocity and its frequency.
- (C) Its velocity and its wavelength.
- (D) Its frequency and wavelength.

2.4.3.10 The diagram shows two light rays in air incident on, and travelling through a slab of glass.

The positions of entry and exit of the rays are shown.

Where is refraction occurring?

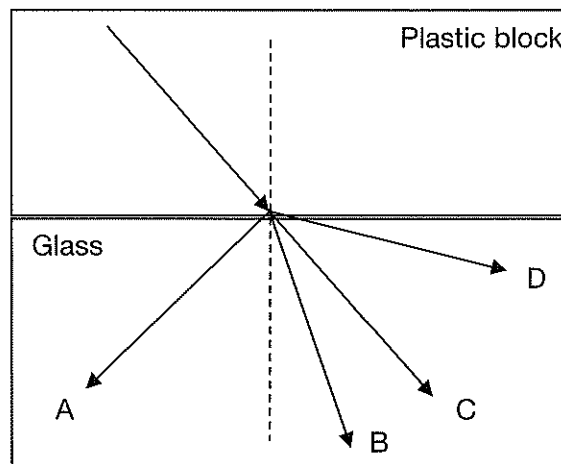
- (A) At W and Y.
- (B) At Y and Z.
- (C) At X and Z.
- (D) At W, X, Y and Z.

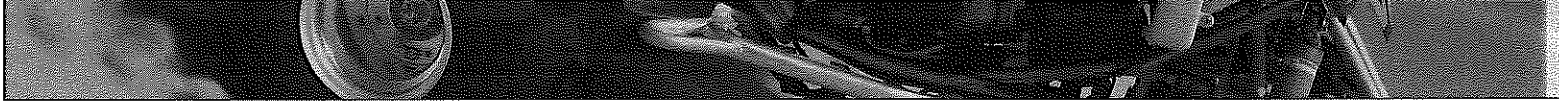


2.4.3.11 The diagram represents a ray of monochromatic light passing from plastic into a block of more dense glass.

Which path will the ray follow in the plastic?

- (A) A
- (B) B
- (C) C
- (D) D

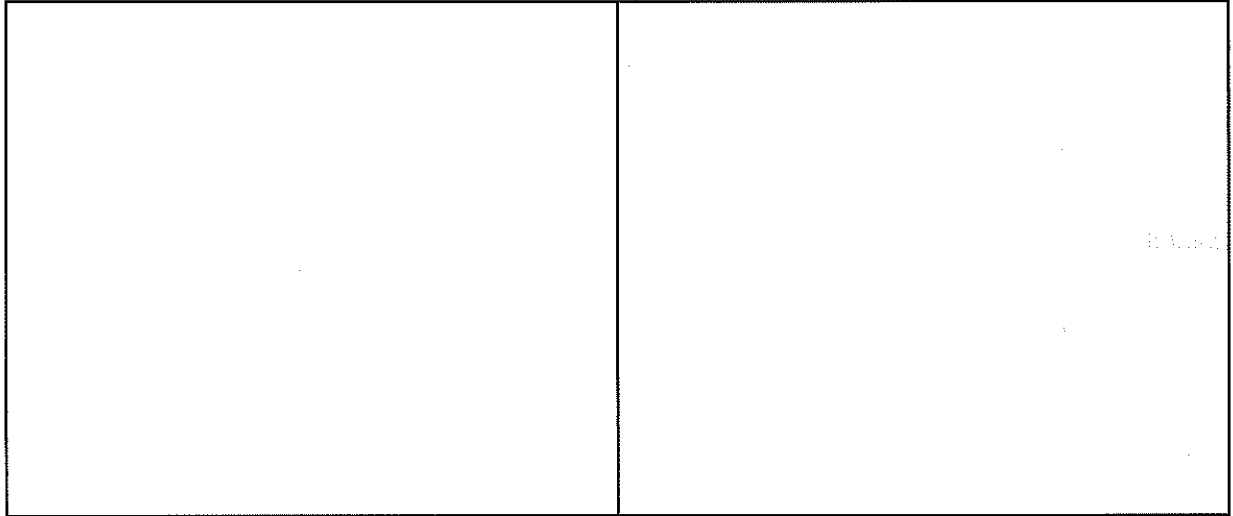




2.4.4 Refraction 2.

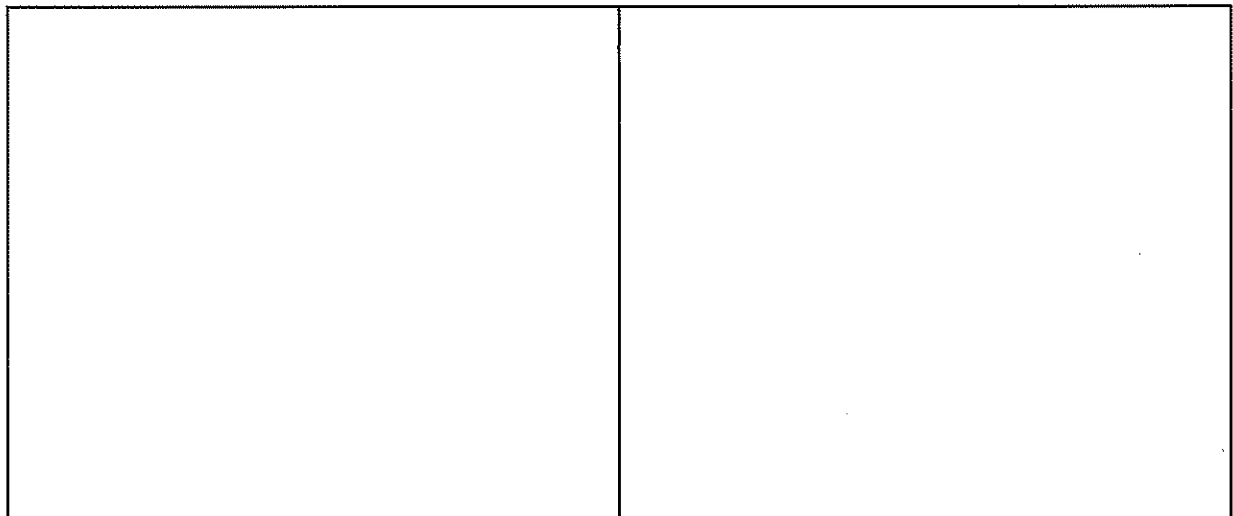
2.4.4.1 Consider the following diagram which shows a region of less dense medium having a boundary with a more dense medium.

- (A) Draw in a ray to represent the refraction of a wave travelling from left to right and from a less dense medium into a more dense medium. Show the wavefronts on the ray.



- (B) On your diagram label:
- (a) The incident ray.
 - (b) The incident angle.
 - (c) The normal to the boundary.
 - (d) The less dense medium.
 - (e) The more dense medium.
 - (f) Wavefronts.
 - (g) Wave moving faster.
 - (h) Wave moving slower.
 - (i) Refracted wave.
 - (j) Angle of refraction.

- (C) Repeat this process for the following diagram which represents a more dense medium and a less dense medium but in the reverse orientation.



- 2.4.4.2** The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise more ideas about the refraction that occurs when electromagnetic waves travel from one medium into another. Some may be used more than once.

Completed sentences	Second halves (not in order)
When electromagnetic radiation passes from less dense medium to more dense medium:	towards the normal.
(a) The wavefront	away from normal.
(b) The wavelength	smaller than angle of refraction.
(c) Refraction is	greater than angle of refraction.
(d) Angle of incidence is	as usual.
When electromagnetic radiation passes from dense medium to less dense medium:	speeds up.
(e) The wavefront	slows down.
(f) The wavelength	will <i>not</i> change.
(g) Refraction is	Increases.
(h) Angle of incidence is	Decreases.
When entering any medium at 90° (i.e. the incident angle is zero):	
(i) Refraction will occur	
(j) The direction of travel	

- 2.4.4.3** Soundwaves, water waves and compression waves in rocks refract under different conditions to electromagnetic waves. Identify the changes in the media involved which cause these waves to refract.

Wave type	What causes refraction?	Identify the specific change
Sound	Changes in density of the media	
Water	Changes in depth of the water	
Earthquake compression	Changes in density of the rock	



2.4.5 Refraction 3.

2.4.5.1

- (a) Define absolute refractive index (ARI).

.....

- (b) Define relative refractive index (RRI).

.....

- (c) Give the equation we can use to find the relative refractive index of the boundary between two media from the values of their absolute refractive indices.

.....

- (d) What is the relationship between the relative refractive index of, for example, an air/glass boundary, and the same glass/air boundary?

.....

.....

- (e) Recall Snell's law of refraction.

.....

.....

- (f) Complete the following sentences to summarise an idea about refraction.

If medium 1 is less dense than medium 2 then the RRI of the boundary is

If medium 1 is more dense than medium 2 then the RRI of the boundary is

- (g) Recall the extended versions of Snell's law of refraction, identifying all symbols used in any equations you give.

.....

.....

.....

.....

.....

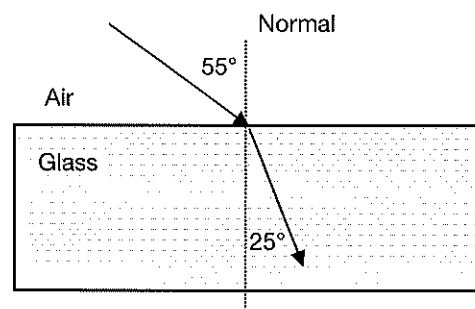
.....

- 2.4.5.2 Use the data in the first table to complete the missing data in it, and the second table.

Medium	Absolute refractive index	Velocity of light in this medium (m s^{-1})	Medium	Absolute refractive index	Velocity of light in this medium (m s^{-1})
Vacuum	1.00	3×10^8	PET plastic	1.58	
Air	1.00	3×10^8	Bromine	1.66	
Water ice	1.31		Flint glass	1.65	
Water	1.33		Sapphire	1.77	
Amber	1.55		Diamond	2.42	

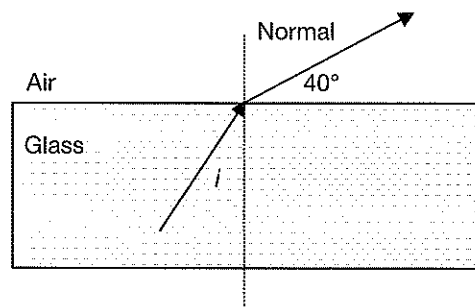
For light travelling from:	The relative refractive index will be:
Vacuum to Earth's atmosphere	A =
Air to flint glass	B =
Air to sapphire	C =
Amber to air	D =
Diamond to sapphire	E =
Water to water ice	F =
Water ice to water	G =
Plastic to bromine	H =

2.4.5.3 The path of a ray of light passing from air into glass is shown in the diagram.



- Find the relative refractive index of the boundary.
- Find the absolute refractive index of the glass.
- Find the speed of the light in the glass.

2.4.5.4 The path of a ray of light passing from the same glass as in the previous question into air is shown in the diagram.



- Find the relative refractive index of the boundary.
- Find the angle of incidence of the light.

2.4.6 Refraction 4.

2.4.6.1 The diagram represents an electromagnetic wave travelling across a boundary from medium 1 into medium 2.

(a) Which is the denser medium? Justify your answer.

.....

.....

.....

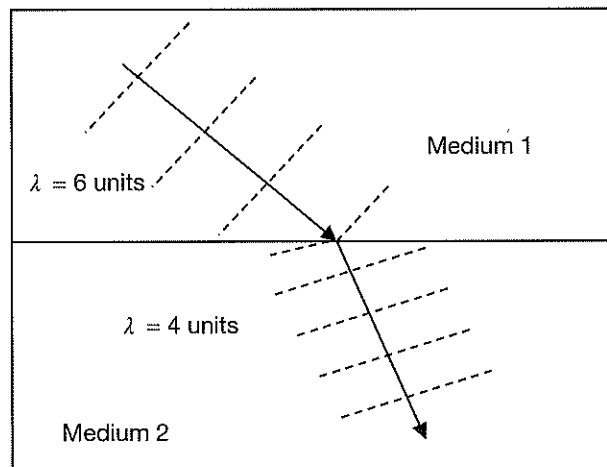
(b) Find the ratio of the frequency of the wave in medium 1 to its frequency in medium 2.

.....

(c) Find the ratio of the wavelength of the wave in medium 1 to its wavelength in medium 2.

(d) Find the ratio of the speed of the wave in mediums 1 and 2.

(e) Find the refractive index of the boundary between mediums 1 and 2.



2.4.6.2 The diagram represents a ray of light travelling from air into plastic, absolute refractive index 1.15, and then into glass, absolute refractive index 1.4.

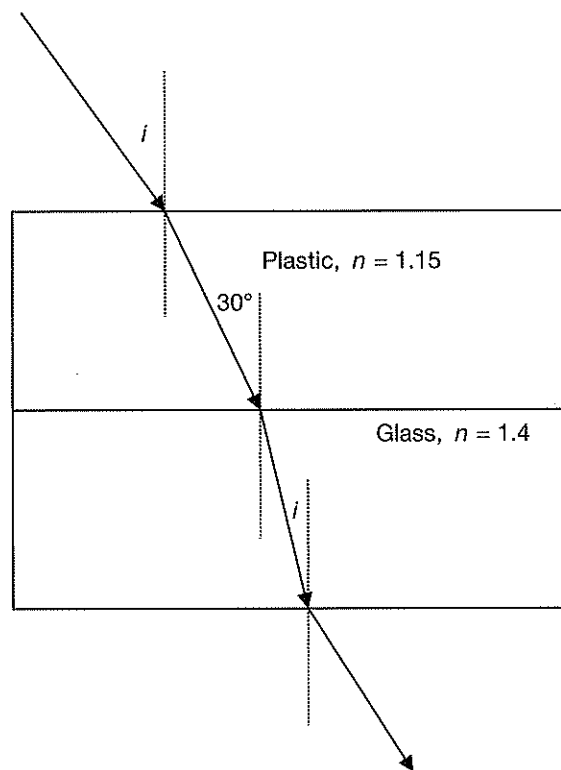
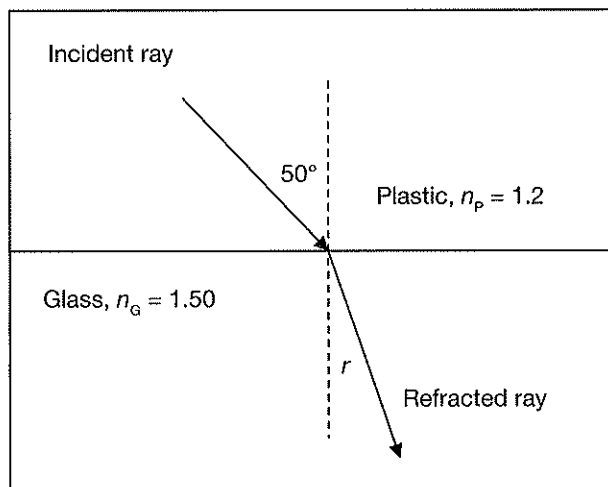
Calculate (to nearest degree) values for the two labelled incident angles.

.....

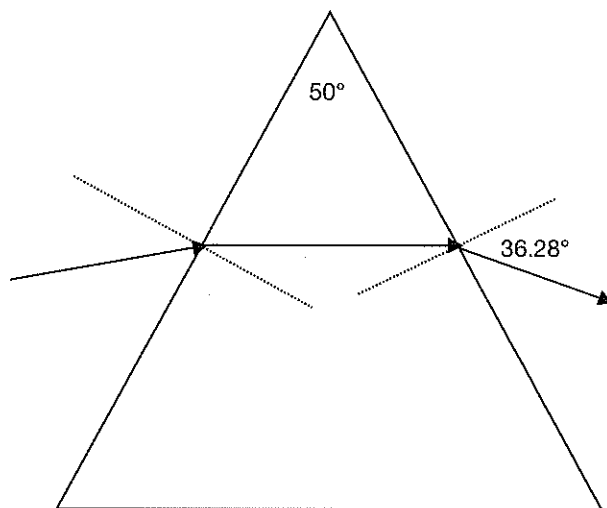
.....

.....

2.4.6.3 Determine the angle of refraction for a ray of light passing from plastic into glass as represented by the diagram.



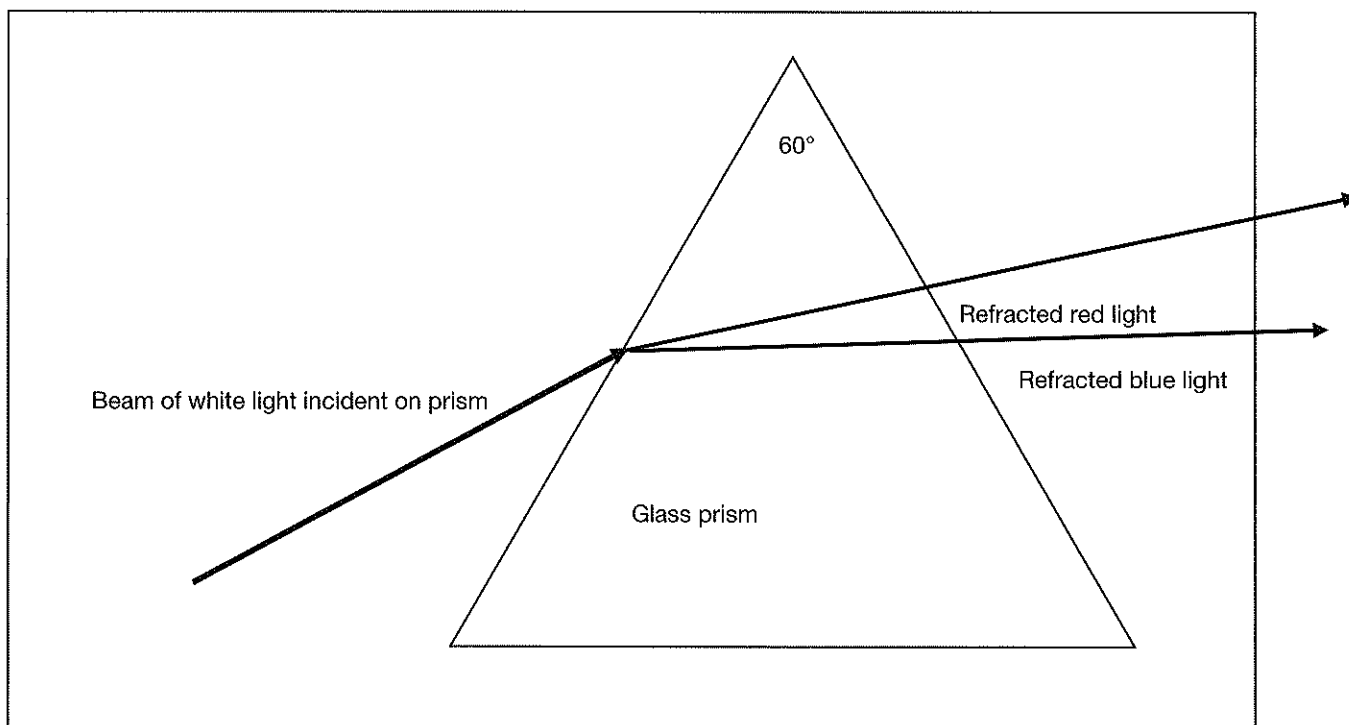
- 2.6.4.4** The diagram represents a ray of light passing through an isosceles, 50° triangular, glass prism. The refracted ray into the prism is parallel to the prism base.



- (a) Determine the angle of incidence of the light onto the left hand side of the prism.
 (b) Calculate the absolute refractive index of the glass.

2.6.4.5 Make appropriate additions to the diagram below, and measurements, so that you can calculate:

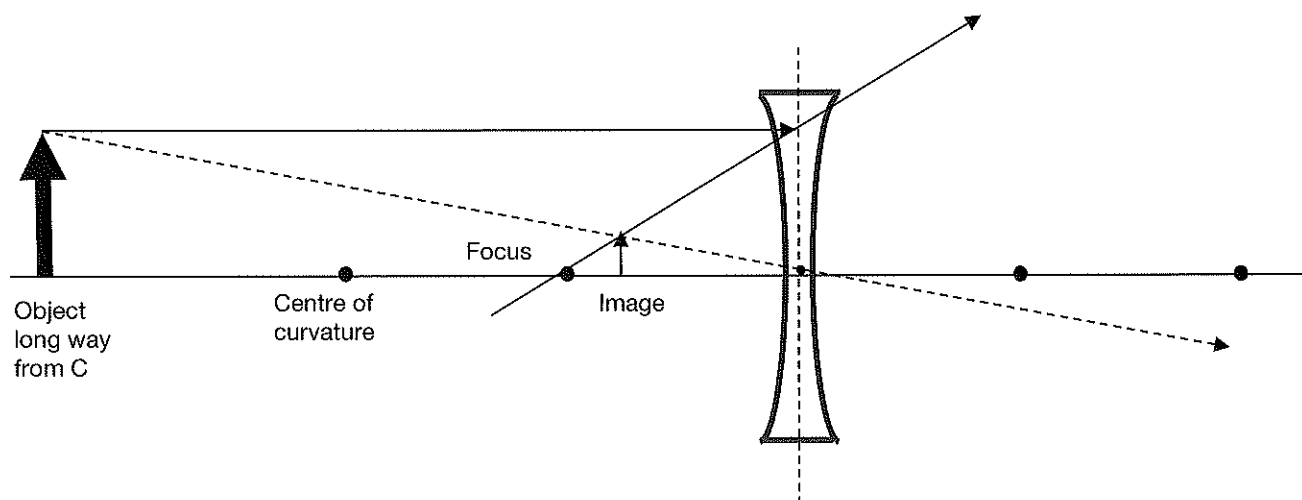
- (a) The refractive index for red light entering the glass prism from the air.
 (b) The refractive index for blue light entering the glass prism from the air.
 (c) Account for the difference.



2.4.7 Images formed by concave lenses.

2.4.7.1 The diagram below shows the formation of an image by a concave lens and the ray diagram we draw to find the position and nature of that image. Note that in these examples double concave lenses have been used and in this case, we consider the refraction as occurring at the vertical centre line of the lens.

In doing this, the diagram represents the two main ideas we implement when drawing ray diagrams for concave lenses.



Examine the diagram, then complete the two sentences to summarise the two ideas we use when drawing ray diagrams for concave lenses.

(a) A ray from the object travelling parallel to the principal axis will refract

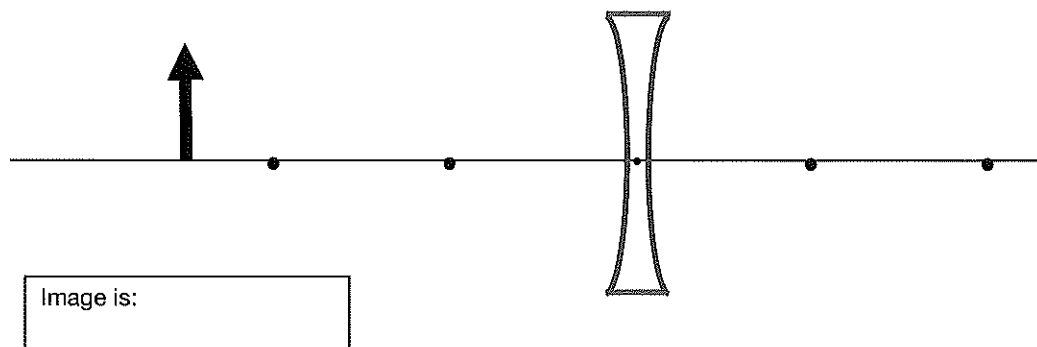
(b) A ray from the object that passes through the centre of the lens

2.4.7.2 Complete the information in the table to describe the possible nature of images formed in concave lenses.

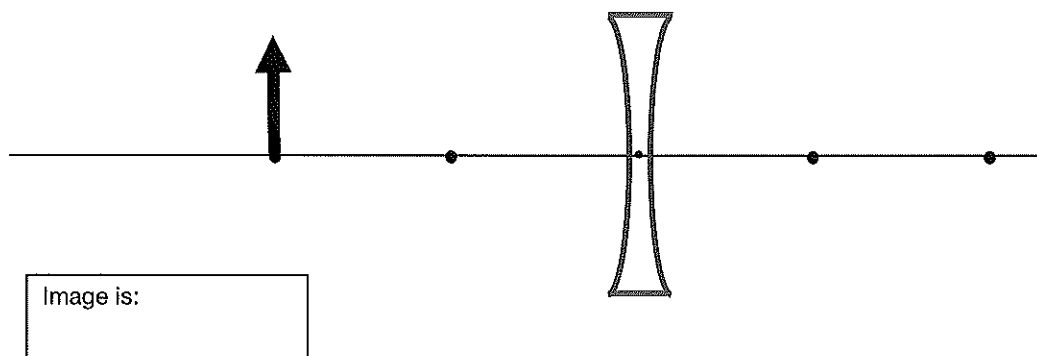
	Image is	Description of this nature
(a)	Real if	
(b)	Virtual if	
(c)	Upright if	
(d)	Inverted if	
(e)	Diminished if	
(f)	Diminished if	
(g)	Same size if	

2.4.7.3 Draw ray diagrams to determine the nature of the images formed by the concave lenses for each of the following positions of an image. Describe the nature of each image.

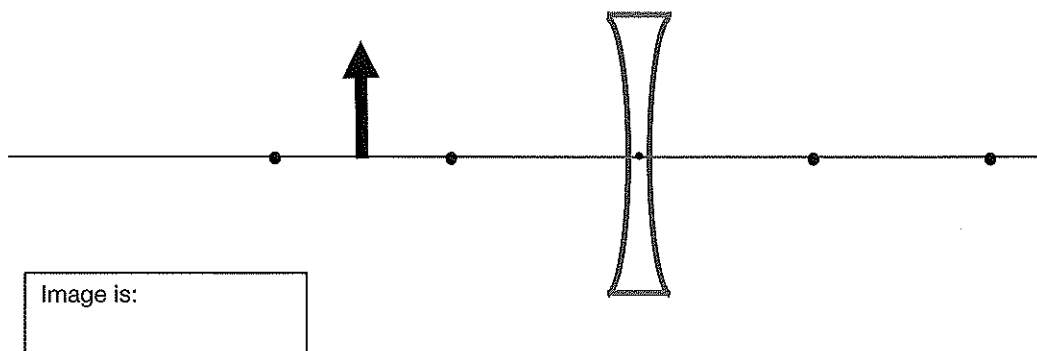
A Object closer to C.

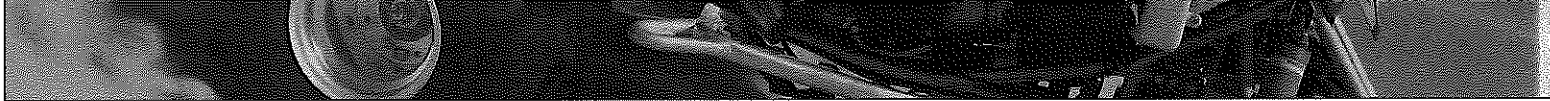


B Object at C.

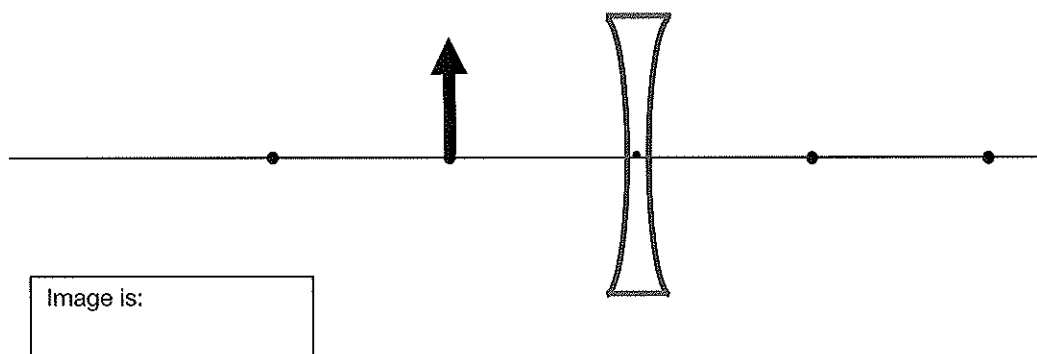


C Object between C and F.

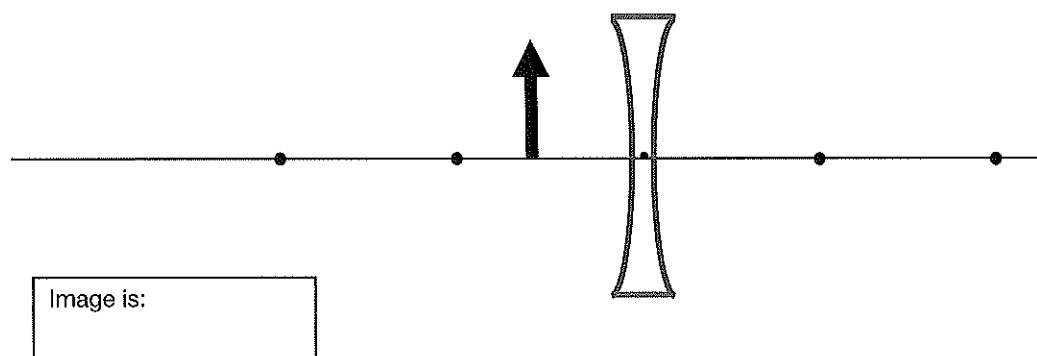




D Object at F.



E Object between F and the lens.



2.4.7.4 According to your work above, a reasonable conclusion that could be made about images formed from concave lenses is that they are:

- (A) Usually real, upright and diminished. (B) Always real, upright and diminished.
(C) Usually virtual, upright and diminished. (D) Always virtual, upright and diminished.

2.4.7.5 Through which part of a concave lens does a beam of light pass without being refracted?

- (A) The focus.
(B) The centre of curvature.
(C) The middle of the lens.
(D) No part. The beam will always refract through the lens.

2.4.7.6 Through which part of a concave lens does light pass without being deviated?

- (A) The focus.
(B) The centre of curvature.
(C) The middle of the lens.
(D) The beam will always deviate from its path as it passes through the lens.

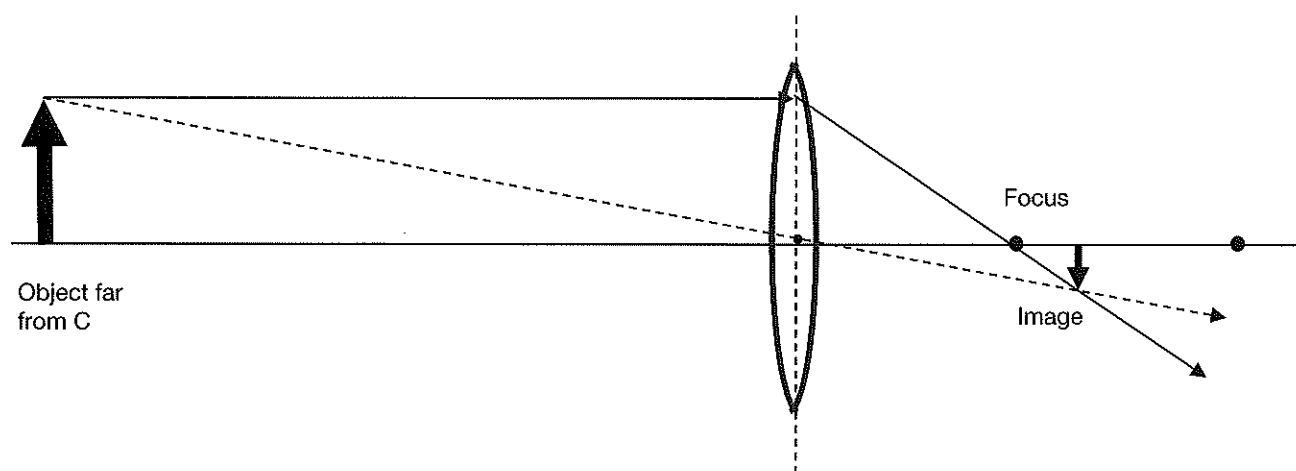
2.4.7.7 Which statement about real and virtual images is correct?

- (A) A real image can be projected onto a screen.
(B) A virtual image can be projected onto a screen.
(C) A real image is on the same side of the lens as the object.
(D) A virtual image is on the opposite side of the lens to the object.

2.4.8 Images formed by convex lenses.

2.4.8.1 The diagram below shows the formation of an image by a convex lens and the ray diagram we draw to find the position and nature of that image. Note that in these examples double convex lenses have been used and in this case, we consider the refraction as occurring at the vertical centre line of the lens.

In doing this, the diagram represents the two main ideas we implement when drawing ray diagrams for convex lenses.

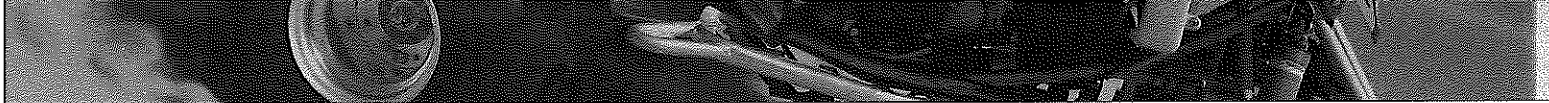


Examine the diagram, then complete the two sentences to summarise the two ideas we use when drawing ray diagrams for convex lenses.

- (a) A ray from the object travelling parallel to the principal axis will refract
- (b) A ray from the object that passes through the centre of the lens

2.4.8.2 Complete the information in the table to describe the possible natures of images formed in convex lenses.

	Image is	Description of this nature
(a)	Real if	
(b)	Virtual if	
(c)	Upright if	
(d)	Inverted if	
(e)	Diminished if	
(f)	Diminished if	
(g)	Same size if	



2.4.8.3 Draw ray diagrams to determine the nature of the images formed by the convex lenses for each of the following positions of an image. Describe the nature of each image.

A Object closer to C.

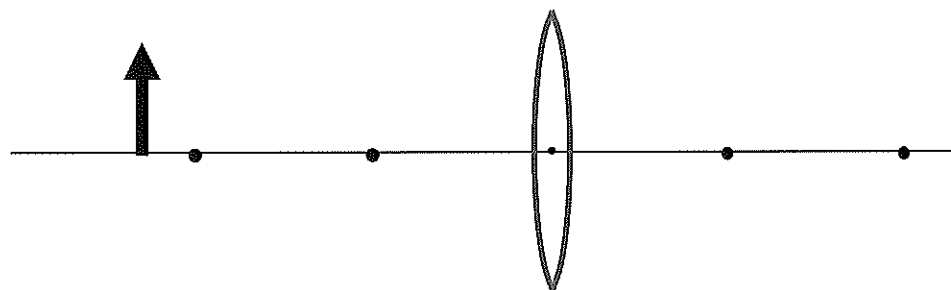


Image is:

B Object at C.

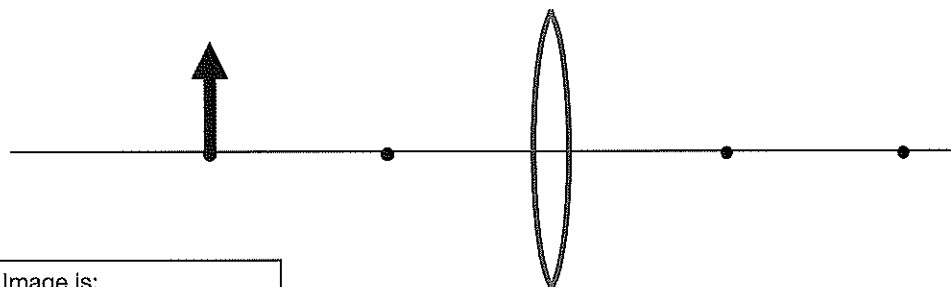


Image is:

C Object between C and F.

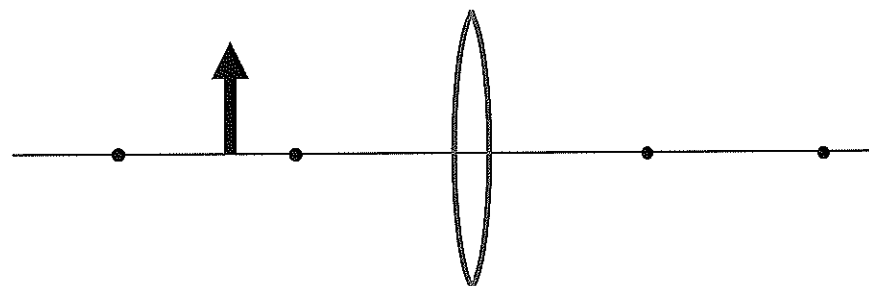
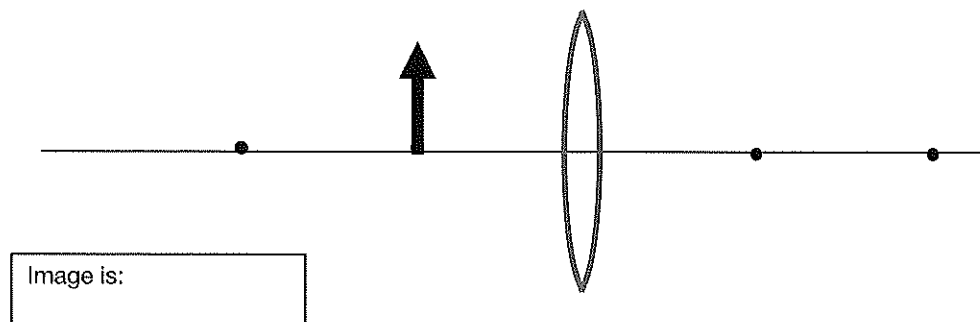
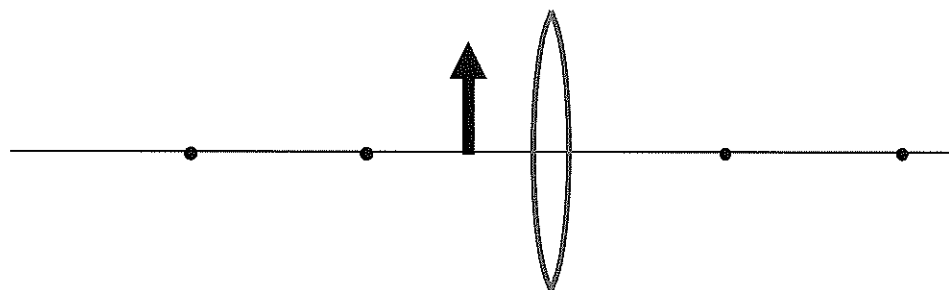


Image is:

D Object at F.



E Object between F and the lens.



2.5.1.1

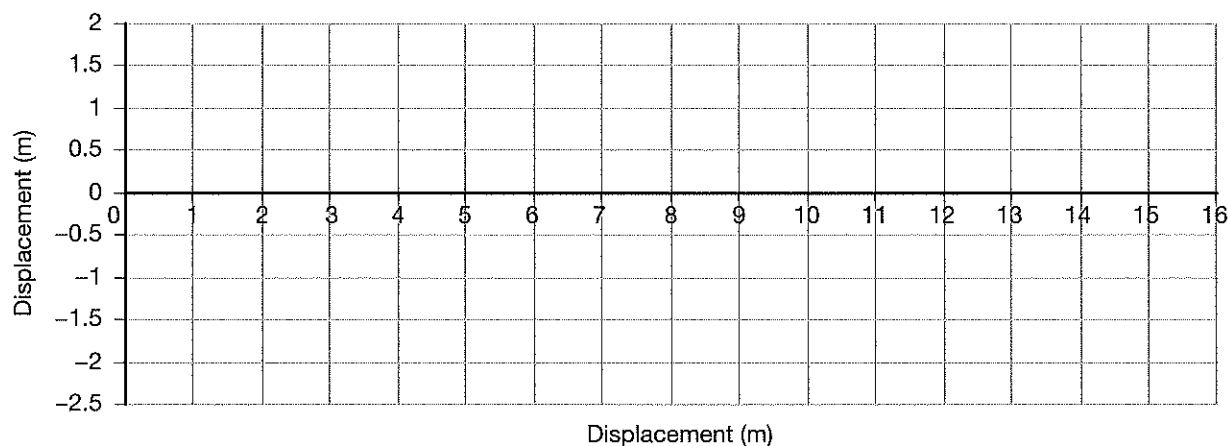
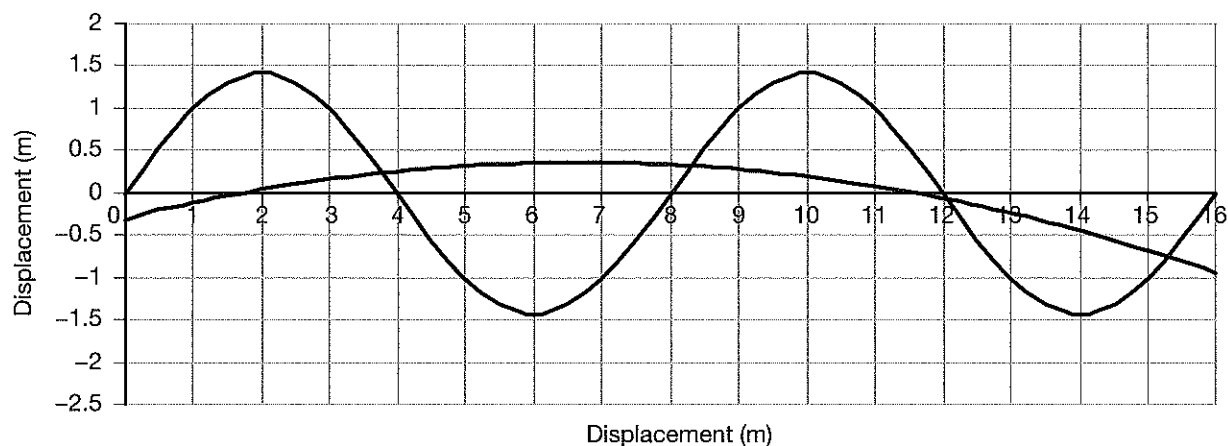
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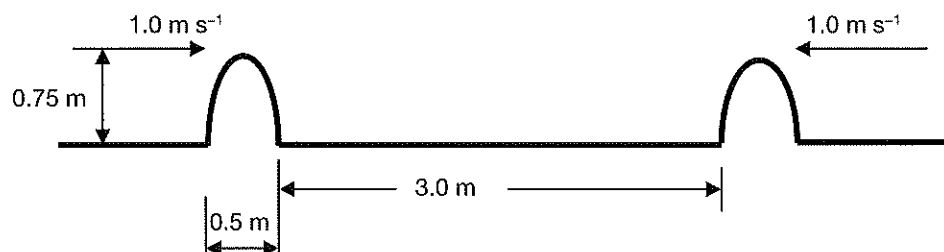
Constructive interference	Destructive interference

- 2.5.1.3** The diagram shows two waves drawn on the same set of axes, and a clean set of axes. Superimpose the two graphs and draw the resultant on the clean axes.



- 2.5.1.4** Consider the wave pulses in the string shown. They approach each other, each moving at 1.0 m s^{-1} . At the instant shown, their leading edges are 3.0 m apart. The wavelength of each pulse is 1.0 m and each pulse has an amplitude of 0.75 m .

On the same diagram, draw what the resultant wave in the string would look like after exactly 1.75 s .





2.5.2 Standing waves in strings.

2.5.2.1

- (a) Define a standing wave.
-
-
- (b) What name is given to point in a standing wave that appears to be stationary?
- (c) What name is given to points of maximum particle displacement in a standing wave?

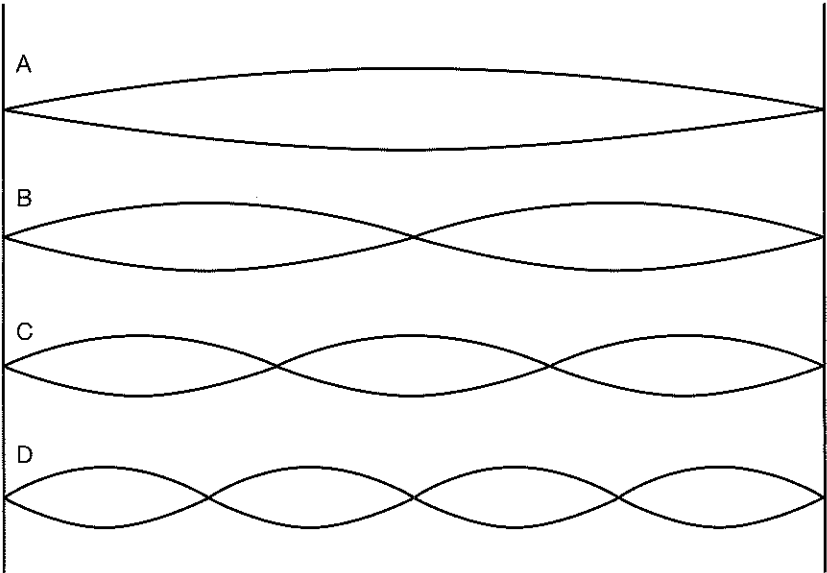
2.5.2.2

- (a) What is the relationship between the fundamental vibration in a string and the first harmonic vibration?
-
- (b) What is the relationship between the harmonic of the standing wave in a string and the frequency of vibration of the standing wave?
-

2.5.2.3 Use the diagram of standing waves in strings to complete the table.

Each string in the diagram is 0.24 m long.

Take the speed of sound in air as 340 m s^{-1} .



Wave	Harmonic	Number of nodes	Number of antinodes	Wavelength (m)	Frequency (Hz)
A					
B					
C					
D					

2.5.3 Standing waves in pipes.

2.5.3.1 The last column in the table gives the second halves of the sentences. Choose the correct second half for each sentence and write it in the first column of the table to summarise ideas about standing waves in pipes.

Completed sentences	Second halves (not in order)
(a) There are two types of pipes in which standing waves occur –	antinode at an open end.
(b) These pipes, like stretched strings, form the basis of	is four times the length of the pipe.
(c) When standing waves form in pipes or tubes, a node will always be found	is twice the length of the pipe.
(d) A pipe which supports a standing wave will always have an	have odd numbered harmonics.
(e) The lowest frequency standing wave that can be formed is again known as the first	open and closed.
(f) In an open pipe, the wavelength of the fundamental frequency	many musical instruments.
(g) In a closed pipe, the wavelength of the fundamental frequency	or fundamental harmonic.
(h) Because of this, closed pipes only	at a closed end.

2.5.3.2 A pipe open at both ends has a length of 0.75 m. If the speed of sound in air is 340 m s^{-1} , find:

- (a) The wavelength of the sound that would produce the fundamental vibration in the pipe.
- (b) The frequency of the fundamental vibration.

2.5.3.3 A pipe closed at one end is 45.0 cm long and air in it vibrates in the third harmonic mode when a tuning fork is struck above it. Find:

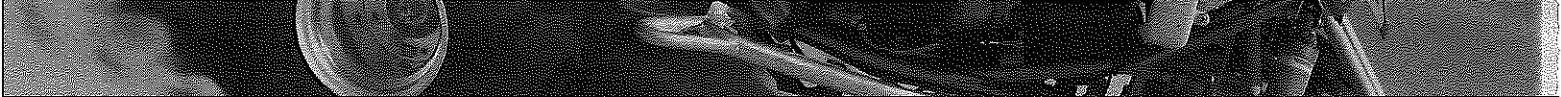
- (a) The wavelength of the sound from the tuning fork.
- (b) The frequency of the sound if the speed of sound is 340 m s^{-1}

2.5.3.4 A pipe, closed at one end has a length of 0.4 m. If the speed of sound in air is 340 m s^{-1} :

- (a) Find the wavelength of the fundamental vibration in the pipe.
- (b) Find the frequency of the fundamental vibration.

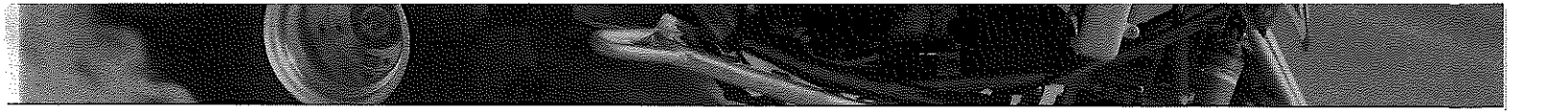
2.5.3.5 A 512 Hz tuning fork is sounded and held near one end of a glass tube which is open at both ends. If the speed of sound is 340 m s^{-1} , how long is the tube if it sets up the third harmonic vibration?

2.5.3.6 A pipe is closed at one end. A tuning fork of frequency 256 Hz is held over the end and sets up the fifth harmonic in the pipe. If the speed of sound is 340 m s^{-1} , how long is the pipe?



2.5.3.7 The open pipes shown below are all 0.6 m in length. Use the diagrams and the information above to complete the table. Take the velocity of sound in the air as 340 m s^{-1} .

2.5.3.8 The closed pipes shown below are all 0.6 m in length. Complete the diagrams by drawing in the shape of the standing waves produced in the pipes, starting with the fundamental in the first pipe and then the next four harmonics in succession. Use your diagrams and the information above to complete the table.



2.6 Mechanical waves.

2.6.1 Mechanical waves.

2.6.1.1 What is mechanical resonance?

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2.6.1.2 Give a common example often used to illustrate mechanical resonance.

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2.6.1.3 You may have experienced mechanical resonance at home when playing your sound system (especially if the subwoofer is on). Briefly describe the effects that occur and are noticeable in this situation.

.....

2.6.1.4 Mechanical resonance is the reason troops of soldiers often march across a bridge out of step. Explain the reasoning for this.

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2.6.1.5

(a) Describe how the sound box of an acoustic guitar is a device for a mechanical resonance and how this assists the player.

.....

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(b) Describe how many musical instruments work by mechanical resonance.

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2.7 Particle, electromagnetic and quantum theories of light.

2.7.1 The particle theory of light.

2.7.1.1 What does the particle theory of light propose?

2.7.1.2 Who was the first modern scientist to propose a particle theory for light? When?

2.7.1.3 On what two observations about light did this scientist base his proposal that light was a particle wave?

2.7.1.4 What experimental evidence did this scientist have to support his proposal?

2.7.1.5 How does the particle model for light explain reflection of light?

2.7.1.6 How did the particle model explain the refraction of light?

2.7.1.7 There is one property of light that can only be explained if light is considered to be a particle and not a wave. What is this property?

2.7.1.8 How do we rationalise this with the fact that all other properties of light can be explained by the wave model?

2.7.2 The electromagnetic wave theory of light.

2.7.2.1 What does the electromagnetic wave theory of light propose?

2.7.2.2 Who was the first scientist to propose an electromagnetic theory for light? When?

2.7.2.3 What experimental evidence did this scientist have to support his proposal?

2.7.2.4 Which other scientist developed mathematical equations that predicted electromagnetic radiation would, if it existed, travel at the speed of light? When?

2.7.2.5 Which experimental discovery soon followed, confirming the prediction of a possible continuous spectrum of electromagnetic radiation? Be sure to mention the scientist involved and the year of the discovery.

2.7.2.6

(a) What property of light, unable to be explained by either the particle or wave theory, was the electromagnetic theory able to explain?

(b) Define this property of light.

(c) Sketch a diagram in the space provided to illustrate your definition.

2.7.2.7 What were the two main observations that the electromagnetic theory could not explain, and therefore decreased its acceptance by other scientists at the time?

2.7.3 The quantum theory of light.

2.7.3.1 Who was the scientist who first proposed quantum theory as a model for electromagnetic radiations? When?

2.7.3.2 What experimental observations did this scientist make which led to his developing this theory?

2.7.3.3 Outline the quantum theory of light.

2.7.3.4 Quantum theory was initially not well supported by other scientists. What happened (include who and when) to change their minds and start to take quantum theory seriously?

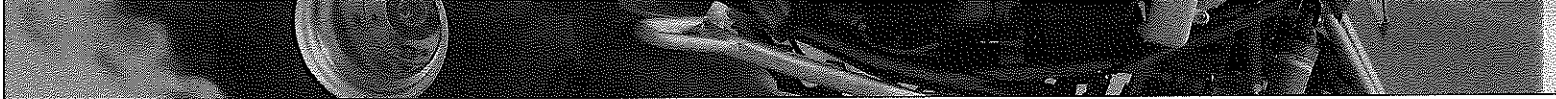
2.7.3.5 Like the other models, quantum theory had its weakness when applied to the nature of light. What was this and how was it integrated into other existing ideas about light?

2.7.3.6 According to the quantum theory, what determines the energy carried by a photon of light?

2.7.3.7

(a) What is the photoelectric effect?

(b) What name is given to the minimum frequency of a photon necessary to cause the photoelectric effect in a medium?



2.8 Advances in science understanding.

2.8.1 Acoustics.

2.8.1.1

(a) What is acoustics?

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(b) Outline the purpose of acoustics.

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2.8.1.2 Outline the benefits of applying the principles of acoustics in a movie theatre or concert hall.

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2.8.1.3 Outline the function of and the way acoustic barriers along the sides of busy motorways work.

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2.8.1.4 Explain the role of high density materials like concrete, or wood in acoustics, giving at least one example each of where they would be useful.

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2.8.1.5 Explain the role of low density materials like air, or fibreglass insulation in acoustics, giving at least one example each of where they would be useful.

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2.8.2 Earthquake waves.

2.8.2.1 Complete the table to summarise information about earthquake waves.

Type of wave	Nature of wave	Travels through	Relative speed	Relative destructive power
Primary (P)				
Secondary (S)				
Surface (L)				

2.8.2.2 Explain why the surface waves are more destructive than P or S waves.

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2.8.2.3

(a) Distinguish between the epicentre and the focus of an earthquake.

(b) Explain how the distance of an earthquake epicentre from a seismic station can be determined.

2.8.2.4

(a) What is the 'shadow zone'?

(b) With the aid of a suitable sketch, show how the shadow zone, and a P waves only zone occurs on the Earth during an earthquake.

2.8.3 Tsunamis.

2.8.3.1 What is a tsunami?

2.8.3.2 Most tsunamis are generated in subduction zones. What is a subduction zone?

2.8.3.3 Describe how tsunamis are generated in subduction zones.

2.8.3.4 Describe the nature of the tsunami wave generated by a subduction zone. Justify your answer.

2.8.3.5 List three other possible causes of tsunamis.

2.8.3.6 What often accompanies the generation of a tsunami in a subduction zone? Explain your answer.

2.8.3.7 How might people living in a low lying area protect themselves from tsunamis?



2.8.4 Monitoring earthquakes and tsunamis.

2.8.4.1 Explain how seismic stations can provide early warning of a possible tsunami.

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2.8.4.2 Outline a typical early warning system for tsunamis other than the use of seismic stations.

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2.8.4.3 Outline how satellite GPS monitoring can be used to give early warning of both earthquakes and tsunamis.

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2.8.5 Making buildings earthquake proof.

2.8.5.1 Describe and outline how each of the following strategies can help minimise the damage caused to a building by an earthquake.

(a) Damping and isolation.

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(b) Reducing resonance.

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(c) Energy dissipating devices.

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(d) Building configuration.

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(e) Centre of mass.

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2.9 Developing models and theories.

2.9.1 The Michelson-Morley experiment.

2.9.1.1 What was the debate raging between scientists that led to experiment like the Michelson-Morley experiment?

2.9.1.2

(a) What is the aether?

(b) Why was the aether thought to be necessary?

(c) What were the hypothetical properties of the aether? (List six with a brief statement justifying each property.)

2.9.1.3

(a) What was the purpose of the Michelson-Morley experiment?

(b) Would a positive result in the Michelson-Morley experiment have proven the existence of the aether? Justify your answer.

2.9.1.4

(a) What term do we use to describe the result of the Michelson-Morley experiment?

(b) What results did Michelson and Morley hope to get in their experiment?

(c) Explain what their actual result meant in terms of being able to make a conclusion from the results.

2.9.1.5

(a) What is the name given to the device Michelson and Morley developed to measure their results in the experiment?

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(b) Why was this device rested on a large block of granite which was floating in a larger tank of mercury?

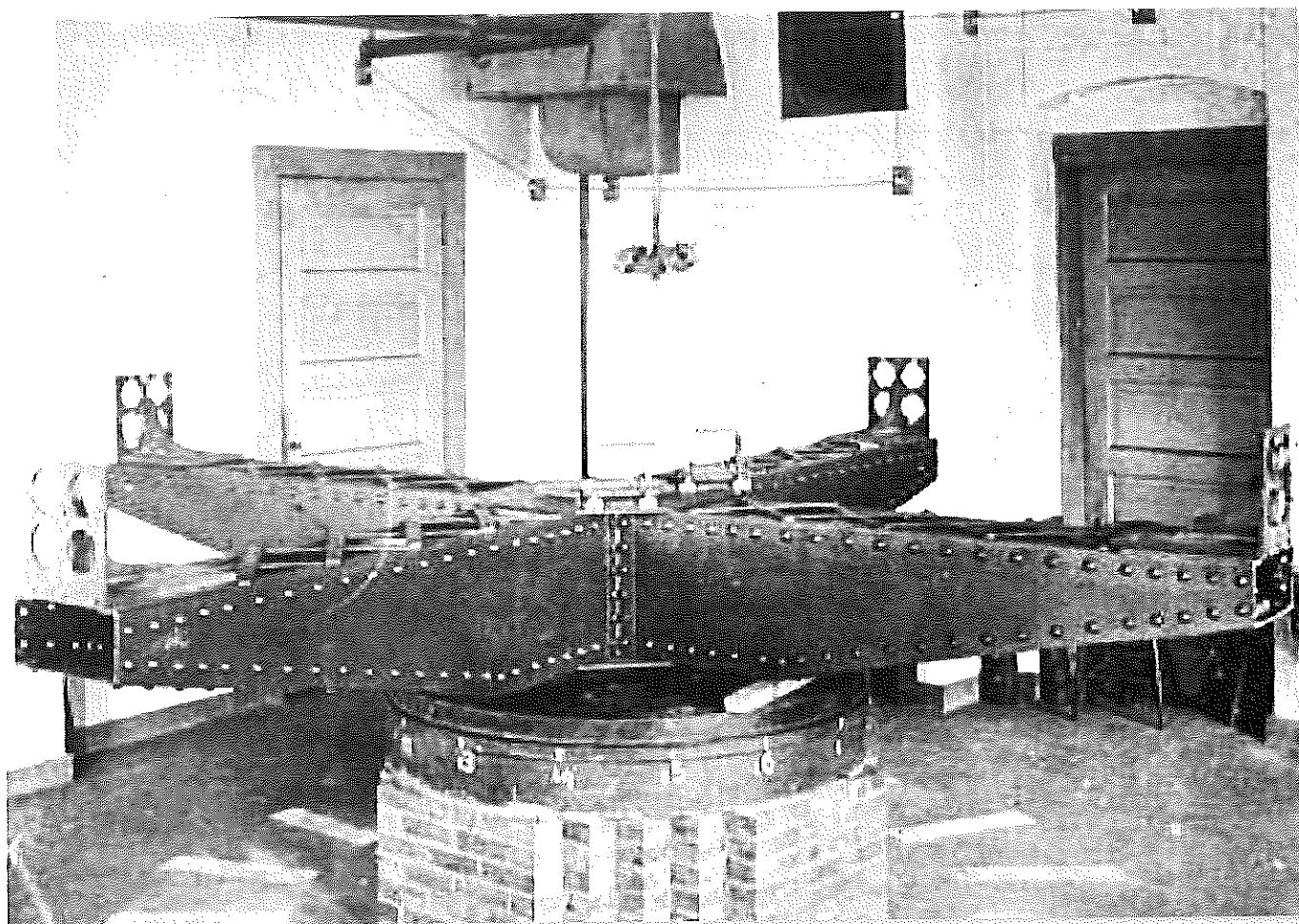
.....

(c) Outline the principles of physics involved in obtaining a positive result in the Michelson-Morley experiment.

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Morley's interferometer was used to attempt to find the aether in 1905.

2.10 The speed of light and the inverse square law.

2.10.1 The inverse square law 1.

2.10.1.1 State the inverse square law for electromagnetic radiation.

2.10.1.2

(a) How does the inverse square relationship for electromagnetic radiation inhibit our ability to study the Universe?

(b) How, in the past 50 years or so, have astronomers been trying to overcome this limitation?

(c) How, in the near future, are astronomers attempting to overcome this problem?

(d) The limitation also affects communications on Earth. How has technology dealt with this limitation?

2.10.1.3 The intensity of sound 60 km from the source is 36 units. Calculate the theoretical intensity heard at the following distances.

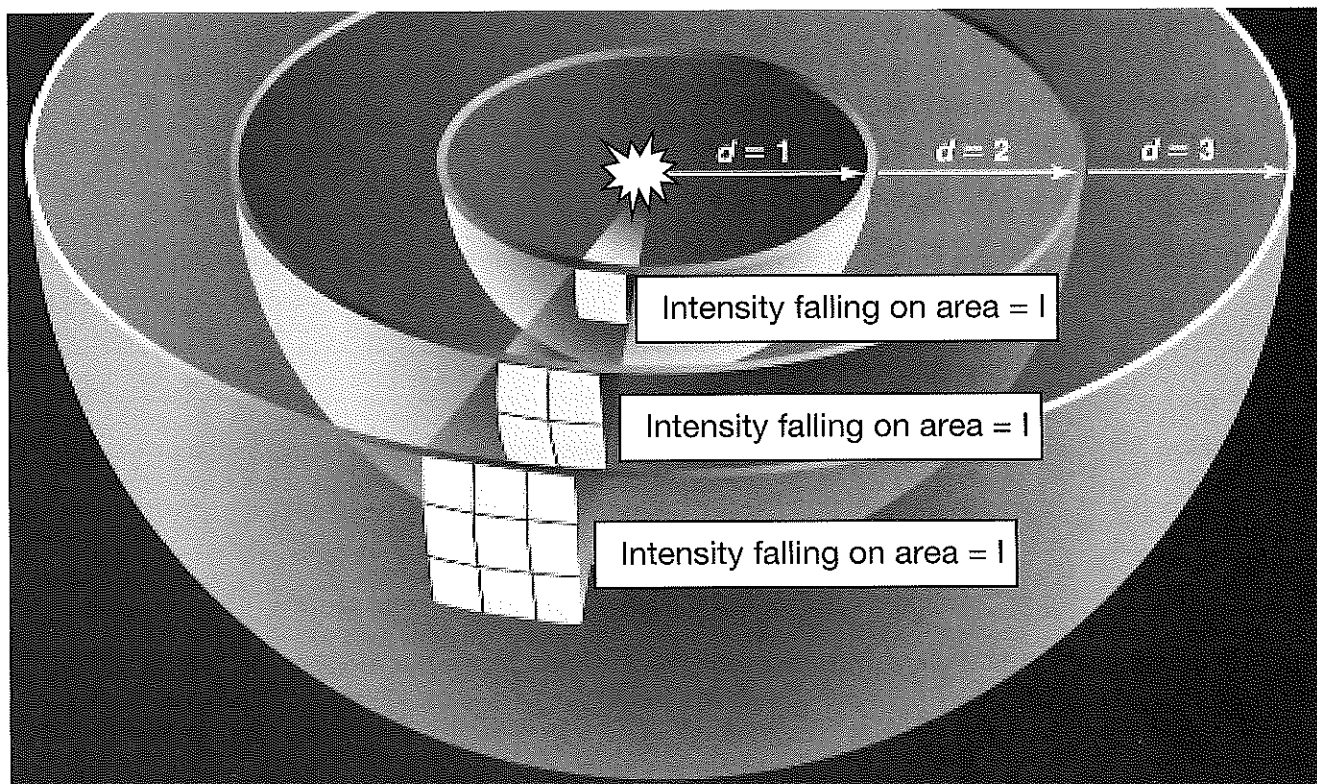
- (a) 180 km
- (b) 120 km
- (c) 90 km
- (d) 40 km
- (e) 30 km
- (f) 15 km

2.10.1.4 The intensity of the light 20 m from the source is 60 units. Calculate the distances from the source where the intensity is as follows.

- (a) 1 unit
- (b) 5 units
- (c) 30 units
- (d) 45 units
- (e) 90 units
- (f) 120 units

2.10.2 The inverse square law 2.

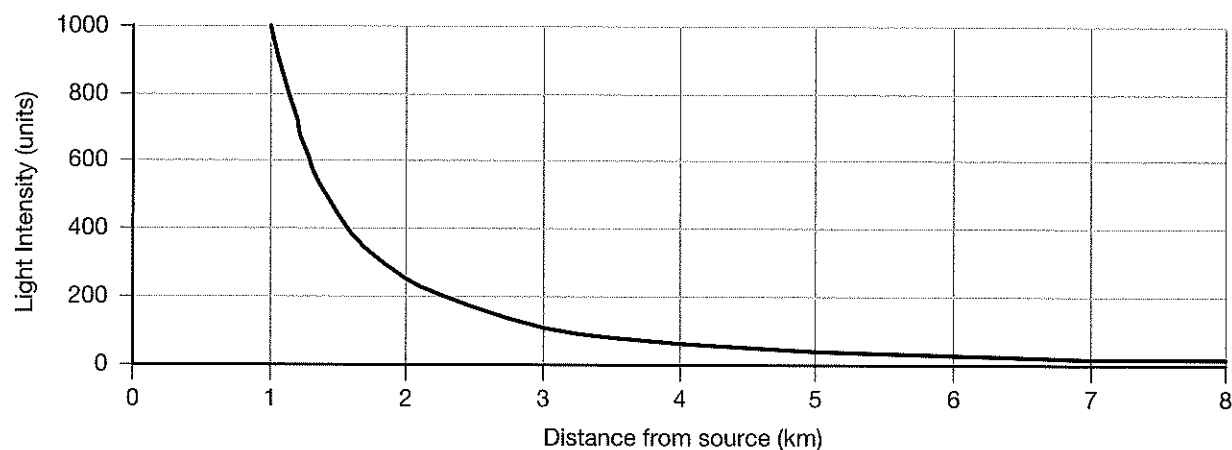
2.10.2.1 Explain how the diagram illustrates the inverse square law for light.



2.10.2.2 Classify each of the following statements as true (T) or false (F). If false, write in the correct statement.

	Statement	True or false?	If false, correction
(a)	If the intensity of a light source is doubled, the intensity at each distance from the source will increase by a factor of $\sqrt{2}$.		
(b)	If the distance from a light source is tripled, then the intensity received will be reduced by a factor of 9.		
(c)	The intensity at a distance of 4 from a light source will be one quarter the intensity at a distance of 2.		
(d)	The intensity at a distance of 3 from a light source will be half the intensity at a distance of 6 from double the light source.		

2.10.2.3 Consider the following graph.



- (a) Complete the data table used to construct this graph.

- (b) If you compare your answer to (a) with the answer in the back of the book, you will see that the author actually included a distance plot for 1.5 km as well as the integral distances. Suggest a reason why this was done.

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- (c) By using three plot points, prove that this graph represents the inverse square law for electromagnetic radiation.

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- (d) Research to identify three other phenomena in science that follow an inverse square relationship.

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